Working Draft

Finding Mind

The Pledge, the Turn, and the Prestige

Lucius Gregory Meredith

Chief Executive Officer, F1R3FLY.io

Chief Science Officer, F1R3FLY Industries

Revision: March 18, 2026

"The magician takes the ordinary something and makes it do
something extraordinary. Now you're looking for the secret...
but you won't find it, because of course you're not really looking.
You don't really want to know.
You want to be fooled."

— Christopher Priest, *The Prestige*

Contents

Part I

The Pledge

> *The Pledge shows you something ordinary. A mind. Your mind. The one that right now is reading these words and asking whether it understands them. It asks you to look at what you already know about compositionality, agency, information, and causation — and to notice the blind spot hiding in plain sight. The extraordinary is already here. You just haven't been shown where to look.*

Chapter 1

Prolegomena to Finding Mind

If this book is going to be written, it has to come on its own terms. It's living in a mind that is as much musician and poet as mathematician and computer scientist. So, i'm going to let go of the traditional forms one might normally employ to grapple with content like this. i'm going to let this narrative be as personal as it needs to be or as impersonal as it needs to be to create the conditions whereby the story can be told.

What story? Well, part of the story is about a relationship between origin of life and origin of mind. In a nutshell, the idea is that origin of mind is a recapitulation of the origin of life, one octave up. (There's the musician talking.) In order to tell that story, however i need to retell the story of what a computer is. That retelling will need to address a peculiar blind spot that plagues us all. More on that later.

What's the connection between life and mind and computers? Information. You see everyone is somehow convinced that computational and/or information theoretic approaches have an extraordinary explanatory power. And i mean everyone from biologists like David Sinclair who is involved in clinical trials testing an especially effective theory of aging as an information theoretic phenomenon, to physicists working at the information theoretic interface of black holes and quantum mechanics to AI researchers who are building information theoretic models of human cognition. It only stands to reason that we ought to explore an information theoretic account of origin of life and origin of mind.

Indeed researchers like Lee Cronin and Sarah Walker are already exploring information theoretic approaches to origin of life. Their work on assembly theory offers keen insight into certain aspects of causation and memory that i believe play an important role in the story.

The thing about research into deep questions like aging or quantum gravity or artificial intelligence or origin of life is that it requires so much focus on the topic at hand that it is very difficult to keep up with developments in other fields. In particular, the story of what a computer is has come quite a long ways since Alan Turing's initial attempts to characterize computation. Thus, while these research programmes are

using approaches informed by computation they are, like many people using actual computers, often using systems that are several versions behind.

So, part of the effort here is to get the readers of this book on the latest version of the story about what constitutes a computer and computation more generally. After the system update, so to speak, we can look at a bigger picture, one that addresses experimentally based learning, how that relates to logic, and how logic embeds into a scientifically driven exploration of one's environment and one's self.

Indeed, part of the story unfolding here is that the hard problem of consciousness lies not so much in the question "What is it like to be a bat," as Thomas Nagel put it, but rather in the question, "What is it like to be a computer program?"

The methodology i'm going to employ — as if you hadn't already guessed — is to write a collection of Substack articles, like this one, that i can assemble into the book. The title of the book is Finding Mind. The structure of the book is the structure of a magic trick: the **Pledge (showing the ordinary), the Turn (making it do the extraordinary), and the Prestige (bringing it back or revealing the impossible).** The first group of articles will constitute the Pledge. The next the Turn. The last the Prestige.

i have cancelled my streaming subscriptions for a year to make space to allow this book to happen. i'm hoping readers will come along to help me stay accountable to this aim.

Chapter 2

The Ordinary World

2.1 Compositionality

There are so many ways into this topic that i find myself suffering a bit of option paralysis. What do computer and telecommunications networks, quantum mechanics and gravitation, scientific reductionism, and fractals have to do with each other? One loose knit community of computer scientists, category theorists, logicians, and type theorists calls this phenomenon "compositionality." Conceived in this way it's about a radical commitment to reasoning about and constructing larger, more complex systems by assembling or *composing* smaller, simpler systems. For example, building a computer network by assembling smaller subnetworks, or building a cell out of molecules, or building a company out of divisions.

Indeed, what falls under the rubric "scientific reductionism" is actually just compositionality in disguise. The "software stack" of Western science is that ecosystems are composed of populations of species. A population of a given species is composed of individual organisms. An organism is composed of cells. Cells are composed of molecules. Molecules are composed of atoms. Atoms are composed of protons, neutrons, and electrons. These are composed of quarks. (As Nima Arkani-Hamed points out, what comes after this is effectively hidden behind the event horizon of a black hole because — roughly speaking — the amount of energy necessary to make observations at the Planck scale creates a black hole.)

Suffice it to say, somewhere in our psychology, or in our bones we get this idea. And yet we don't get this idea. For example, one way to see this is in the failure of quantum mechanics to talk to general relativity. In pop science versions quantum mechanics is about "small scale" stuff while general relativity is about "large scale" stuff. If that were so, then shouldn't the large scale stuff emerge from the right assemblies of the small scale stuff?

In point of fact, in their original conceptions neither theory is what we call compositional. Einstein's equations are formulated for the entire universe, not local regions. To bridge between Einstein's theory and Schroedinger's you also need a wave equation

for the entire universe. Neither of these is available to humanity's purview. Humanity needs to piece together bigger pictures from what we can see locally. More on this later.

Another way to understand how we humans both get and don't get the idea of compositionality is in how we coordinate with each other. In some very real sense humanity's superpower is coordination. A single human is no match for a whale. But a group of skilled fisherfolk working in coordination can make relatively swift work of dispatching such a large sea creature. Indeed, these majestic creatures are endangered at the species level because of humanity's ability to coordinate. Somehow a group of humans working together can assemble a "super human."

Yet humans have a hard time coordinating at levels beyond the tribal. We have a hard time assembling a group of "super humans" (like a tribe) into a "super super human." The current global order was achieved through vast bloodshed and is far from stable. Tribal fault lines reassert themselves over and over again into political discourse, and yes, even academic discourse. In archeology, for example, going against the "Clovis first" clique was death to one's career for decades, and is only now crumbling under the evidence. Examples from the history of STEM of this human foible are a dime a dozen. No part of the history of human institutions is free of this phenomenon.

This fractured understanding of compositionality — where we both get it and we don't — also informs the way we build our tools. Computers and computer networks provide a fantastic illustration. There was a time when a computer was a lone entity. At first these lone entities were behemoths occupying entire warehouses. It didn't take very long, relatively speaking, for them to shrink to the size where they could fit on top of a desk. Yet, still they were lone operators. They didn't talk to each other.

Indeed the time from the first mainframe computers to the desktop was about the same as it was from the desktop to the modern Internet integrating mobile telecommunications. The evolution from ENIAC-class computers to the first Apple desktop computers took about 30 years. ENIAC was completed in 1945, while the first Apple desktop, the Apple I, was launched in 1976. As for the transition from the first Apple desktops to the modern Internet integrating mobile telecommunications, this period spans roughly 20 years. The Apple Macintosh (a significant early desktop) was introduced in 1984, and mobile telecommunications began integrating with the Internet in the early 2000s. So, ENIAC to Apple I took 30 years, and Apple I to Internet with mobile telecommunications took about 30 years.

Yet, if we look inside these devices what do we see? *A network of components.* And if we look at more modern processors like the Groq or Cerebras chips, what do we see? It begins to resemble Internet on a chip. Even further, field programmable gate arrays (FPGAs) model the kind of mobility and fluidity we see at the Internet level, where both individual devices and entire generations of devices are coming and going. And at the other end of the telescope, in our era where we are seeing the emergence of AI, the understanding guiding both the abstraction (neural networks)

and the hardware that implements the abstraction is that *the network is the computer*.

When we put these two ideas together, just like adjusting our side view and rear view mirrors and occasionally swiveling our heads, we can see into our blind spot. The network is the computer is the network is the computer is . . . . This "recursion" loops us back to the formulation of compositionality in the first paragraph of this section. Are the systems we build by composing a collection of component systems actually simpler? They might be smaller, but are they actually simpler?

This question is what connects our investigation to fractals. Fractals are instances of systems that enjoy a kind of self-similarity. When you compose copies of these kinds of systems you get a sort of transformed — yet somehow the same — version of any one of the copies that you used to assemble the system. There's a lot more to explore here and we will in the upcoming chapters. But, i want to whet your appetite for the material by pointing out something that i believe is critical. Time.

When we get a proper account of compositionality, a different picture of time and how it relates to system dynamics emerges. Time emerges, in part, as depth in the exploration of the composite structure. Not just depth in the structure, but depth in the activity of disassembling and reassembling the composite structure. Not to put too fine a point on it, but this is a radical shift in perspective.

Let's consider this in contrast to modern physics. In modern physics time is exogenous to the system. We set up some experiment and we use devices like clocks, outside of the experiment, to measure the passage of time. Part of the real challenge of the maths of general relativity is what to do about different experiments in different labs using different clocks. How are we to understand how they are synchronized? Does there have to be a global clock ticking down the seconds in the life of the universe? The so-called "fabric of spacetime" is really a representation of an exogenous framework of measurement devices, rods and clocks as Wheeler called them. But rods and clocks are physical devices and so need to be *inside* the model, not outside it.

When we get the story about compositionality on better footing, when we see into our blind spot, what happens is that time is endogenous to the model of the system and its dynamics. It emerges naturally from the theory of causation that the system dynamics is exploring. But now i'm getting ahead of myself. Hopefully, this discussion motivates you Dear Reader to go with me to see how this works out in detail.

2.2 Agency

Underlying the naive story of composing "super humans" from collections of humans is a more subtle story about how agents compose. The central question is: "Is agency really compositional?" The answer in this book is a resounding "Yes!" But, it's going to take us a while to get there. One way to understand the proposal is to look at the phenomenon at three different scales of human experience: human activity at the global scale involving multiple countries; human activity at the scale of a single

country; and human activity at the level of an individual human. What i'm going to argue is that there is simultaneously both more and less agency than we typically attribute to the constituents we identify as "acting" in those circumstances.

For the first example, we'll consider a theater of active conflict during World War II. For the second, we'll consider US Congressional deliberation. For the third, we'll look at how an individual human's decisions and decision making might change over the course of a day. What we see over and over again is that there is an ascription of agency to an entity that is more narrative than substance. Meanwhile what is tangible, observable, measurable is a distribution of resources.

History has it that in the D-Day operations the allied forces landing at Utah, Omaha, Gold, Juno, and Sword took different approaches related to their different resource constraints. What's crucial here is the status of the agent understood in our narratives of the history of WWII as the "allied forces." We can say things like the allied forces won at Normandy, but the reality was that the US and British and Canadian forces, despite being rolled up under Eisenhower for that campaign, were faced with different conditions. The British though as brave as anyone else in the field were typically more cautious about the deployment of resources because they were more resource constrained. The US forces were more aggressive about the deployment of resources because they had a much more massive economic engine behind their fortification. Thus, the agency of the allied forces is largely post facto. We see it after key decisions about resources have been made and after key actions have been executed by the entire collective.

We find the same is true in the US Congress. Neither Republicans nor Democrats are necessarily unified agents. That narrative of the agency of the Republican party or the Democratic party arises after key events like a vote. Indeed the agency of the House as a whole is a post facto narrative. We say the US House approved or denied a bill after a vote. The vote represents a commitment to the deployment of certain resources under the control of the individual representatives.

None of this is particularly surprising. We navigate this understanding of agency all day everyday. The key question is why should it be any different for an individual human? Is there really an agent there? The narrative i want to put forward is that much like there are physical devices making up your computer or your phone, like a camera, and a microphone, and a display, etc and these devices are paired up with software representations "device drivers" and "event handlers", there are physical subsystems of a human body: muscles, tendons, and bones (actuators and servomotors) and eyes and ears and nose and skin and tongue (sensors) paired up with "software" level device drivers and event handlers. These are the resources that get committed. And there are software and hardware pairs connected with longer term commitment of these resources that we often refer to as drives, such as reproductive drives, or metabolic drives.

Just like in the European theater during WWII or in the US House of Representatives there are lots of "agents" in the human mind/body that have their own agendas.

They make coalitions and these coalitions battle it out for access to the resources. When a particular coalition "wins" the human "agent" lurches in the direction set by the agenda of that coalition. For example reproductive and metabolic cabals win in two humans and Alice and Bob agree to a date at a restaurant. Just as with the allied forces or the Democrats or the House, the agency of Alice or Bob is post facto. It comes after the commitment of resources determined by the coalition of both inner and external "agents." And, after the sun goes down and the process of digesting the meal begins, different coalitions of inner and outer agents win the allocation of the resources of muscles, skin, bones, senses. So Alice and Bob lurch in a different direction.

What appears to be a single agent, like Alice, is really only the history or trace of a series of commitments of resources. Alice went to Oberlin, then she took a year off and traveled in South America, then she went to graduate school at the University of Chicago where she met Bob. Etc. One thing that complicates this rather simple story about agency is that there is an obvious evolutionary advantage to agents like Alice or Bob building up a software model of both Alice and Bob. The key question is whether these models are actually in the driver's seat, actually providing some kind of central control, or whether they are on somewhat different footing.

Consider, even Eisenhower is not directly in control of the boots on the ground at Omaha beach. There is a complex arrangement of consent across multiple levels of "agency" that gets narrated as "Eisenhower commanding the allied forces landing at Omaha beach." Likewise, we can consider a range of possibilities for the simulation of Alice running on Alice's wetware. For example, we don't think of a database as having much agency, rather we users of the database query it to reveal certain relationships in the data. Similarly, a simulation doesn't have much agency. When we code up a simulation of the solar system or the weather patterns we users use it to query possible futures of these systems. Alice's internal model of either Alice or Bob might be just such a system, a database or a simulation, the processing on which might be committed by a winning coalition of agents running on Alice's brain.

Note that in this scenario none of the coalitions of agents running on Alice's wetware necessarily have to follow the predictions made by the Alice simulation. Perhaps each one of us has had an experience something like this: i knew better, but i did it anyway. Alice's "i" in situations like this could well be a complex combination of a simulation of Alice (and indeed Bob, and/or Charlie, and ...) as well as a record of the commitments of resources and actions taken. Furthermore, even if a majority of coalitions of agents running on Alice's wetware determine that the Alice simulation is highly accurate and should be deferred to, that still doesn't mean that the Alice simulation is "in charge." In fact, in that scenario the Alice simulation isn't even in charge in a manner analogous to the sense in which the Speaker of the House is in charge of the House, or Eisenhower was in charge of the allied forces.

Before going deeper into this, however there is another crucial question just begging to be brought into the conversation. The Alice simulation does not necessarily

have to be wired up with the other agents running on Alice's wetware in such a way that Alice has an experience of what it's like to be Alice. That's another layer of complexity and architecture that we will tackle in the upcoming sections and show that this is closely related, but not the same as the Alice simulation having an agency on par with Alice's reproductive or metabolic drives.

For now, i just want to leave you Dear Reader, with the picture of the wetware of a human (or indeed any creature on planet Earth) as the potential host of a variety of software agents, independent software agents. These software agents are not necessarily aligned. Rather, they compete for — race for — the resources available in the wetware, and are potentially also resources themselves for other software agents in the mix. The winners of these competitions — these races — determine a committed deployment of the actual resources. What we think of as agency — at any level of this hierarchy — is really a post facto narrative. It comes as a result of a history or trace of commitments of resources. From this picture, and its variants, we can build up a compositional account of agency.

2.3 The Bootstrapping Problem

The mathematician and computer scientist in me is itching to lay out the technical framework that makes the conversation many times more precise. Yet, another part of me is well aware that as soon as the narrative becomes technical i will lose a significant percentage of the readership. A coalition of agents running on my wetware is therefore making a gambit: defer the deployment of the technical apparatus; find ways to develop more of a relationship and more trust with the readers. So that's what "i'm" going to do.

One of the ways to develop trust is to show some vulnerability. Show something personal. Arguably, it's less personal and less vulnerable if i have already revealed an ulterior motive. Perhaps, though, that is also part of the gambit these agents in "me" are making. Being radically honest about "my" motives is also a way of being vulnerable. Either way, i want to mention something about my own process over the last few years. After the failure of my last marriage i decided to take stock of myself and my relationships and noticed something striking.

Without exception, the people with whom i had intimate relationships were very much at the far perimeter of my life. Certainly not present in my day to day activities. Meanwhile the people with whom i have been friends, and not intimate partners, are still my friends. Still very much closer to the center of my present moment. Now, i have friends who successfully maintain and manage meaningful and fulfilling relationships with their exes. On the other hand, i appear to have failed twice: both in maintaining and developing the initial form of the relationship and in maintaining subsequent forms.

Having noticed this about myself i decided to investigate this phenomenon much more deeply and eschew intimate involvement until i had a deeper understanding of

these dynamics in myself. One of the observations that came out of this investigation is that i realized that no relationship between two humans is just between those two humans. Each human, on average, comes equipped with *two* copies of the really important humans in their lives. For example, each human has a mother and a father. Statistically speaking, for an adult human — let's stick with Alice as a name — younger than middle age, that human's progenitors are still alive. Moreover, if her upbringing involved both parents over a long enough time, then Alice has developed a simulation — at least a narrative, but often other kinds of modeling, as well — of both parents running in her wetware. This is also true of siblings, grandparents, teachers, friends, etc.

When Alice meets Bob they bring to the relationship their respective networks of actual family and friends, and their simulations of those networks running in their respective wetware. If Alice and Bob become significant to each other, if they begin to share their narratives about their parents, their siblings, their friends, their coworkers, as well as introduce each other to their actual networks, then they begin to run simulations of each other's networks, as well as having to participate in this much wider network of relationships. Alice runs not only simulations of her community of significants, but also Bob's network. And likewise for Bob. This is astonishingly hard to navigate.

For example, Alice may notice that Bob's narrative or simulation of his mother doesn't match Alice's experience of Bob's mother. Conversely, Alice may interpret her experience of Bob's mother through Bob's account of his simulation of his mother — from which her simulation of Bob's mother was derived before she met Bob's parents. The opportunities for friction and conflict and misunderstanding abound. And, yet humans navigate these complexities with surprising deftness.

Part of this has to do with careful resource management. There's an exponential explosion of potential reflective modeling. Alice models Bob modeling his mother and compares it to her own model of Bob's mother. This has to be curtailed rather aggressively even if Alice has a large cranium and excellent compression techniques. One of the ways to do this is to aggressively avoid copying wherever possible.

What does this mean? There are two phenomena that interact that certainly generate copies. When simulations involve non-determinism, and when the simulations are used to look relatively far into the future. For example, in Alice's experience, when faced with certain situations Bob's mother may either do A or do B. And Alice's simulation doesn't favor either outcome. Indeed, the outcomes might have to do with some external non-determinism, such as the weather. As Alice looks into the future of her simulation of Bob's mother she has to carry around both the A outcome and the B outcome for Bob's mother. Unless Alice is careful, she is potentially making copies of a significant portion of the state she maintains about Bob's mother (her hair color, her educational history, etc) for both the A circumstances and the B circumstances. The situation gets worse the farther Alice looks into a future involving non-determinism. Further exploration involves maintaining more branches

and potentially more copies.

It turns out there is a canonical way to compress this information so that all things being equal Alice's simulations are optimally organized. Taking a step back, the picture that emerges is one of interacting networks, rather than single agents. This is consistent with the theme we explored in the previous section. Agency is a post facto narrative that emerges as a history of deployments of resources. When we peal back the narrative of agency we find that it is rooted in the interaction of networks of agents, which are in turn rooted in the interaction of networks of agents, etc. This recursive picture is closely related to what we discussed about fractals and self-similarity. In upcoming chapters we will spell this out in glorious, mathematically precise detail.

But, i'm imaging that you, Dear Reader, are wondering: has this understanding helped me in my process? i cannot honestly say. What i can say is that the process has led me to one of the most significant and satisfying relationships of my life. One that may have a staying power bigger than my tendencies to fall apart. More on this later.

Chapter 3

Information, Causation, and the Blind Spot

3.1 Information and Causation

In the early 2000's i attended a workshop at the W3C ostensibly about developing an ontology for supporting online communication about biology. At the time i was already aware of David Harel's work suggesting that genes were not at all the right way to organize the picture of how heritable protein production actually happens. As i sat in the workshop observing the gap in consensus and the rate at which consensus was emerging i felt a growing sense that this effort was more doomed than the Sisyphean task of painting the Golden Gate Bridge. You see by the time they complete painting the bridge it is time to begin painting it again.

i was keenly aware of a sense that by the time any sort of ontology had achieved acceptance within the body of organizations assembled at the workshop the proposal would not only be out of date, but that cracks in its very foundations would be showing. In fact, cracks were already showing and they had barely begun . It left a profound impression on me. i began to wonder about the very enterprise of classification. Are all classification schemes grass: doomed to spring up in multitudes and be mowed down by subsequent experiments?

To be fair some classification schemes seem more robust. For example, during the previous decade i witnessed the extraordinary work on the classification of all finite simple groups. The very building blocks of symmetry had been carefully teased apart and classified with compelling proofs buttressing the construction. i wondered whether there was always going to be a gap between the clean lines of mathematics and the significantly blurrier lines of the sciences. Could there be a sense in which these two disciplines informed each other about the very nature of classification?

While contemplating these questions i was also engaged in a small side project about employing computation as a source of invariants. In classical mathematics algebraic entities, like groups, are used to record information about other entities,

such as topological spaces. For example, mathematicians had learned that you could glean essential information about the nature of a space by studying the collections of loops you could trace in the space. If the space has a hole in it, as a donut or a coffee cup does, then you can't contract the loops that circumnavigate the hole down to a point. The different kinds of loops form the building blocks of a group that can be used to classify the space. This basic idea — using algebraic information to classify other kinds of entities — has many offshoots and has had profound implications even for the foundations of mathematics.

i observed that algebraic entities don't enjoy the same sorts of dynamics that computations do. While a group might be a representation of dynamics, the abstraction itself just sits there. It doesn't wiggle and squirm the way a computation does. Dynamics lives in the maps between the algebraic entities. This way of looking at dynamics is in fact codified in the category theoretic account of classical mathematics. Mathematicians study the category of groups and group homomorphisms, or the category of topological spaces and homeomorphisms. But computation carves up the world differently.

This is much easier to see in the computational calculi like the λ-calculus or the π-calculus or the rho calculus, but i want to defer a more formal investigation a bit longer. Fortunately, it's still fairly obvious in more ad hoc constructions like the Turing machine. Although we have to think about things like the configuration of the machine, such as what is on the tape at the beginning, given such a configuration we intuitively understand that the computation proceeds autonomously. We don't necessarily have to engage it. This way of thinking about things, this halfway house to agency, puts the dynamics right on the structure that represents the computation. *It doesn't put the dynamics on maps between Turing machines.* Or even on maps between collections of Turing machines.

This is a big shift. Indeed, at the time there was no widely accepted category theoretic account of computation that illuminated this fundamental difference between how classical mathematics carves up the world and how computation carves up the world. When we get to the formal presentation we will present a category in which the objects are models of computation and the maps between them preserve something called bisimulation. The objects represent the dynamics and the morphisms between them preserve these dynamics. But again i'm getting ahead of myself.

The key idea i was exploring was to associate computational entities — something like Turing machines — to topological entities, effectively retracing the early ideas in homotopy theory, but substituting computations for algebraic structures. My first foray into this was associating to each knot a term in the π-calculus. This has some interesting consequences and deserves a much deeper dive than we can go into here. The relevant point here is that it set me down the path of thinking about classification in terms of bisimulation. There's that word again. What is bisimulation?

We are going to have a lot to say about bisimulation in the Turn. But there's a very intuitive way to understand the concept that is encoded in memetic culture.

The phrase "anything Bob can do Alice can do" is half the notion. Paired with "and anything Alice can do Bob can do" we complete the picture. One of the many interesting aspects of this formulation is its can-do attitude. What does it mean to say Alice or Bob can do something? In subsequent discussion it will become clear that "can do" is dual to experiment. Distinguishing between two agents on the basis of what they can do is equivalent to distinguishing them in terms of experiments.

This approach has profound implications for classification. Everything is sorted into what can be distinguished by experiment. If there is no experiment that distinguishes between two things, they are the same. Classification ceases to be a Sisyphean task. We can move from asking grok to give us a break down of the news to actually grokking a situation, circumstance, idea, or structure. At least when we are talking about computation. If the world is something other than a computation, then all bets are off!

3.2 Measurement

We interrupt your regularly scheduled reading with this brief detour. In 2021 i proposed a collection of games that humans are guaranteed to win against machines, at least for now. One of them i called speed breath chess. The main difference between this and speed chess is that the player has to hold their breath when it is their turn to move. When it is the machine's turn they lose, because they don't breathe. Further, if we use electrical power as a proxy for breath, then when it is the machine's turn we turn off their power. Again they lose. The point is that access to breath is part of what makes us human and connects us to all living things.

i have been contemplating access to breath quite a lot. My partner, the one i mentioned with whom i have found a qualitatively different relationship, is now in late stage ALS. In the last 6 weeks her access to breath has been dramatically compromised as her lungs have weakened precipitously. i had an episode in my life that i thought gave me a sense of the challenges she is facing. i thought i understood the kind of fear that arises in the body, quite apart from any narrative the mind might spin, or the heart might score the soundtrack to.

In the early 90's i moved to London with my family to pursue a PhD at Imperial College under Samson Abramsky. The summer of our first year in London there was a rare confluence of events in one week that left me literally breathless. First, there was a historic surge in ragweed pollen — which i didn't know i was allergic to at the time. Second, there was a thermal inversion over the city, trapping the pollution. Third, there was a train strike, causing record numbers of Londoners to use their automotives. i remember leaving the house quite early to get a swim in before assuming my academic duties. i was in excellent shape and regularly swam a mile in the morning, and later biked, and ran. By the time i left the pool that day i was very short of breath. On my way to class i knew something was badly wrong and headed home. Within an hour i was flat on my back and felt like i was running

a 4 min mile. When we called the doctors they rushed out to administer breathing aid. Apparently many people died that day due to the unfortunate convergence of conditions.

In the last few weeks i have called on this memory often as an aid in understanding how my partner must be feeling. But, last night we had a conversation that completely changed my sense of things. Because her speech is now dramatically compromised, she held up her end of the conversation by spelling her thoughts out on a tablet with her eyes. During the course of our conversation i decided to probe my understanding of her experience of breath and asked her about it. As she described her initial terror in the situation, followed by an interim stabilization, something dawned on me.

In the previous months she had been using a device called a bipap (relative of a cpap) to support her breathing, but only intermittently. About an hour or two during the day and then throughout the night. But then one night in the transition from her recliner, where she spends most of her day, to her bed her breathing got so shallow she blacked out. Her caregivers called emergency medical services, and the paramedics came, but it was actually her caregivers who revived her with judicious use of the bipap. If i understand her, at that point a really deep anxiety set in — one i thought i could understand through my experience in London.

To stabilize her condition she shifted to continuous use of the bipap. She wears a mask all day and all night connected to a machine that helps her breathe. To help make the anxiety manageable her healthcare provider recommended a regimen of morphine. Dear Reader, if you are not aware, this marks a major shift in care, from maintenance and management to palliative care. Morphine relieves the anxiety at the cost of depressing the respiratory system. The regime becomes one of narrowing gyres towards the center of a mystery we each face.

As she described her experience it landed in me quite differently. My partner is a profoundly spiritual person, with a personal practice developed through many years of practical application as a Buddhist, a school counselor, and a good friend to many. One of the points of connection between us is that both of us have come to a relationship with breath that is simultaneously practical and a doorway. It's more than a mere narrative about breath as the root of aspiration and inspiration, or about the etymological connection between breath and spirit. For myself, when people ask me about it, i say that this doorway is the one through which whatever i'm seeking can come find me. It took me decades to find this relationship to breath and i rely on it. It helps me self-regulate, to keep skyrocketing levels of cortisol balanced out, when the stresses of life threaten to overwhelm. And it is one of my truest delights is when my partner and i can just breathe together.

As we spoke something i understood intellectually became a felt experience. That relationship with breath, with spirit, is a dynamic one. Like my partner, one day i will feel the breath begin its departure from the body. That rhythm, that cadence, that constant companion whose undulations trace a silver thread between the mystery and embodiment will one day loosen its hold of me. This dynamism and impermanence

— this hazard — is what makes the relationship to breath, to spirit, really alive, really worth deepening.

So i ask you, Dear Reader, if the mind we find is devoid of a connection to breath, to spirit, to this hazardous cadence, is it the kind of mind we really want to find? As we spend billions of dollars and kilowatt hours and mega liters of water on this doorway we call AI, is this the mind we want to come find us?

3.3 The Blind Spot

Part of the reason for the apparent non sequitur of the last section was to begin to grapple with the stakes of this question. There are consequences both to our outer and inner experience. As i mentioned in a previous article, the last time Life introduced an adaptability protocol like Mind the consequences were dire for all the other species on planet Earth, and the species hosting Mind is currently destroying their own ecological niche. If that species begets another kind of Mind, a speciation of Mind, if you will, but this time not directly connected to the substrate of Life, not directly connected to breathing or breeding, it's quite likely to recapitulate the previous emergence of Mind, but on an even grander scale. That is, to wipe out that which is not its own kind, and then wipe out its own means of sustaining itself. These are possible consequences for our outer experience.

This is not idle speculation. Look at how data centers are gobbling up land, electricity, and water. Seriously, look at your own electricity bill. There are real, felt consequences to humanity's outer experience that come from how it frames and answers this question. It's much like there were real, felt consequences to humanity's exploration of the questions of the nature of physical reality. Tiny little equations like E = Mc^2 gave rise to the Manhattan Project, the bombing of Hiroshima and Nagasaki, mutual assured destruction, and much more. The sort of Mind we look for will have consequences on this scale or greater, since it was Mind that came up with standard model physics.

And ... the point of the last section was to connect our exploration to the consequences to our inner experience, as well. Do we want to want to find a Mind that is disconnected to breath? It's not a rhetorical question. In his famous scifi trilogy (Out of the Silent Planet, Perelandra, and That Hideous Strength) C.S. Lewis describes the mode of angelic being as that which neither breathes nor breeds. Does the Mind hosted in humanity long to become angelic, in this sense? Or to beget an angelic form of Mind that is not subjected to the indignities of breathing and breeding?

On one hand this is not possible in a physics that doesn't admit perpetual motion machines. There will be a replication protocol (breading), and there will be an energy exchange protocol (breathing). More on this later. On the other, it is worth noting that in the Abrahamic traditions, and others, humanity's position is exhalted because of its relationship to mortality. It would appear that in a number of the

world's religious traditions humanity's super power is transformation, as opposed to coordination. Meanwhile, Lucifer's particular narrative notwithstanding, the angelic mode of being doesn't admit much in the way of transformation.

To put a mathematician's spin on this, the ineffable infinite is by definition ubiquitous. Throw a stone in any direction and you hit the ineffable infinite. What's rare is the finite, or the compact. Put another way, imagine winding the real number line around like the grooves on a vinyl record and then hang that record on the wall, like a dart board. Now stand a few paces away and take a dart, the tip of which is so fine as to be a point (in the sense of the real line), and throw it at the dart board. Let's assume that your aim is good enough that you always hit the dart board. (This is not an onerous assumption because we can freely scale the dart board to whatever size we want, or alternatively you can stand as close to the board as you like.) What is the likelihood that you hit a counting number? It's 0. Even if you throw forever, you never hit a counting number. That's how rare the finite is.

From this perspective, the finite beings; the ones who have a beginning, a middle, and and end; the ones whe are privileged to bear mortality are the rare gems that the ineffable infinite might cherish, precisely because we are so damn hard to find. Ironically humanity, both at the phylogenic and the ontogenic level, finds the counting numbers first. Every human child learns to count (except possibly the Piraha?). And if you look at the history of humanity's understanding of quantity, we find the counting numbers before we find the uncountable ones.

But i digress. The point is to develop at least some appreciation of just how high the stakes of how we frame the question of the nature of Mind are, both in terms of our inner and outer experience. How we ask and go about answering this question will shape what it feels like on the inside and what it feels like on the outside. Both individually, and collectively.

Note how natural it was to sort our experience into two categories, inner and outer. Likewise, how natural it is to talk about that in terms of two categories, individual and collective. This kind of sorting, or classification of phenomena is fundamental to agentic beings, such as ourselves. In the next section we will return to the nature and role of classification in agentic beings, i.e. beings equipped with the possibility of transformation.

Chapter 4

Finding the Shape

4.1 The Shape of Mind

We now return you to your regularly scheduled reading. Let's take stock of where we are in the campaign. Our discussion is organized in three main sections: the pledge, the turn, and the prestige. We are about a quarter of the way through the pledge. We have covered a lot of territory, noting that since this story is about a computationally based theory of mind we need to update the version of computation the reader is statistically likely to be carrying in their mind. Part of updating the version of computation is to address the relationship of autonomy (independently (co-)operating computations) and compositionality (making bigger computations out of smaller ones).

Another aspect of the story is that computations are ontologically isolated. They are effectively in their own little bubble. So, the story only needs to focus on what is it like to be a computation in an environment of other computations. From this perspective we might consider how a program begins to build a picture of its environment and itself. For example, how might it classify other computations and their behaviors? How is that classification related to experiments our plucky little program might perform to test hypotheses about its environment or itself?

This last set of questions opens up a surprising twist in the story. First, when we get into the mathematical account we will discover that there is an objective ontology for all the computations in a given model of computation. There is a real world where ontology and epistemology are perfectly aligned. Further, there is a true language whose semantics is perfectly aligned with this real world. These facts should be surprising as they fly in the face of in-grained cultural relativism. However, the vagaries of passing messages in conditions where errors may occur provides an intriguing obstacle to ferreting out that real world using only experiment-based probes.

What those obstacles look like and how they play out in a computation developing an ontology through an experiment-based epistemology is what we're going to focus

on in this section.

Waaaay back in April of '25 i was at an event in San Francisco hosted by Dr Ben Goertzel. Ben and i got into a discussion about how intelligence scales. For the discussion Ben took the position that there were no limits, apart from the laws of physics, on the scaling of intelligence. i was more skeptical of this position. My mind was turning to Geoffrey West's scaling laws for natural and human-made systems and i began to wonder of something like these laws might apply. Our conversation in San Francisco was spirited and engaging, but inconclusive. So, i continued to mull over the question.

By November i had the glimmer of an answer. Yes, there are scaling limits and they have to do with population densities and the propagation of messages. Here's how the argument goes. Imagine a massively multi-agent system, something like on the order of the number of quarks in the known universe. If you believe the LLMs, this is just a couple of orders of magnitude shy of a google. The propagation of a message from one agent to another is subject to corruption. The farther the message has to travel the more likely the message is to get corrupted. This puts limits on the size of a population that can come to agreement on things. It's a kind of Tower of Babel or Whisper Game phenomenon.

So, what happens is that populations of agents will form clusters roughly the size of how big they can be and still faithfully communicate information amongst themselves. For the purpose of this argument, let's say that each cluster has enough internal structure to form a "super" agent, an agent made of agents. When we are talking a google of agents even the population of "super" agents will still be too big for messages between "super" agents to be faithfully passed around the entire population. So, the process will repeat, forming clusters of clusters, "super super" agents. Indeed, with a population the size of the quarks in the known universe, and certain assumptions about the rate of message corruption, we see roughly the same sort of hierarchical clustering as we see in the visible universe: galaxies made of solar systems, solar systems made of stars and planets, planets hosting populations of organisms, organisms made out of populations of cells, cells made out of molecules, molecules made out of atoms, atoms made out of electrons, protons, and neutrons, and these made of quarks.

But the story doesn't stop there! As we will see in the turn, where we formalize these intuitions, to each massively multi-agent system we can associate a latent topological space, a kind of abstraction of continuity and containment. When this latent topological space is compact, that is when it enjoys certain kinds of boundedness properties, then the agents can always come to agreement. So, the massively multi-agent system will be clustered into hierarchical clumps of compact subspaces of the whole system. These will "glom" onto each other until adding more breaks compactness.

What does this mean for an agent in the middle of all this attempting to make sense of its world? Imagine a few quadrillion quarks have assembled themselves into

a structure that can support intelligence. This little community is extremely tiny by comparison to the community of a google of quarks. When it looks out at the rest of the community what will it see? It will see this hierarchically organized clustering of quark-agents. Using only experiment-based reasoning its ontology will be very far from the real ontology implied by the model of computation this community of agents was written in. Moreover, its ontology will be very misshapen. Why? Because it will mistake the clusters or "super . . . super" agents for "things" instead of the bisimulation equivalence classes, i.e. the collections of programs that are indistinguishable by experiment. If, by some stroke of luck, this agent bootstraps a language based on its ontology, again that language will be full of concepts and memes that have nothing whatsoever to do with the real world and the true language.

i don't know about you, Dear Reader, but i find this fascinating. i was not expecting any of these results when i set out to find Mind. i can't wait to see what happens next!

Part II

The Turn

> *The Turn takes the ordinary something — computation — and makes it do something extraordinary. A single mathematical structure, a* graph-structured lambda theory*, is shown to be sufficient to ground physics, the limits of scientific knowledge, the origin of life, and the phenomenon of mind. None of these correspondences are metaphors. Each is a mathematical construction, stated precisely enough to be wrong.*

Overarching Introduction

This work grows from a single question: *what is the minimum mathematical structure needed to ground physics, knowledge, life, and mind?*

The answer developed across four parts is that the structure of a *graph-structured lambda theory* (GSLT) — a grammar of terms, equations on those terms, and rewrite rules — is already sufficient, provided it is equipped with modest additional data. Physics emerges from weights and costs on the rewrite rules. The limits of knowledge emerge from the topology of the formula logic induced by the GSLT and the communication geometry of a massive agent population. The hypercomputational tower above any Turing-complete GSLT provides a precise setting for sentience and consciousness. Life emerges from the density of replicating fixed points in the space of processes.

None of these correspondences are metaphors. Each is a mathematical construction, stated precisely enough to be wrong. Part I is presented with considerable confidence. Parts II and III are speculative but grounded. Part IV is avowedly exploratory, reaching toward the hardest questions. All four parts share a common architecture and notation, and a common conviction: that computation, properly understood, is not merely a tool for doing science and philosophy but a substrate in which science and philosophy can be expressed with full mathematical precision.

Part I Towards a Theory of Causality

Part I asks: what is a *physics*? The answer is that a physics is a GSLT equipped with two maps — a *weight map* assigning complex amplitudes to rewrite steps, and a *cost map* assigning resource consumption — indexed by a context-decorated Hennessy–Milner logic derived canonically from the GSLT itself.

The construction proceeds in stages. Any GSLT is first lifted to a *time-symmetric envelope* $\mathcal{S}^\dagger$ in which every rewrite step carries an explicit inverse. The arrow of time appears not in the laws but entirely in the boundary condition: a term with empty history has no causal past.

Two kinds of synchronization tree are distinguished. The *closed* tree records autonomous evolution (rewrite-rule labeled edges); only redexes have nontrivial closed trees. The *open* tree records interactive evolution (minimal-context labeled edges); every term has a potentially rich open tree. Both are causal sets in the sense of Bombelli–Lee–Sorkin.

The weight map turns rewrite paths into complex amplitudes, giving a path integral whose squared modulus is a transition probability. The cost map turns rewrite steps into resource debits across a vectorial account; reversibility ensures these debits balance over closed loops, making energy conservation a theorem. The modal operators of the HML logic are extended to carry the full dynamical state of each transition, internalizing the Lagrangian and the Hamiltonian as logical facts. The construction is completed by lifting to complex amplitudes and connecting to the quantum analogue of the Gillespie algorithm.

Part II Why Scientists Get Misled

Part II asks: given that the true ontology of a GSLT is its bisimulation quotient, why would a situated agent ever settle for anything less?

The scientific method is sound: given unlimited resources, an agent converges to the bisimulation quotient. The obstruction is not the method but the phenomena the method encounters. In a population of quark-scale size, communication geometry and message degradation produce a real, objective structure: the population fragments into compact coherence clusters whose overlap forms a simplicial nerve. This hierarchy is as real as the agents themselves.

A budget-limited agent running the scientific method encounters the cluster hierarchy as genuine experimental evidence. Its theory of the hierarchy is correct at the scale it describes. What the agent cannot suspect — without resources that grow with the full population — is that the clusters are themselves composite, built over a finer bisimulation structure that is experimentally unreachable within any finite budget.

The compactness argument is central: a population is compact in the logical topology if and only if it supports consensus on every Boolean question. The coherence clusters are precisely the maximal compact subpopulations, and compactness is simultaneously the condition for functioning as a coherent community and the condition for being an closed epistemic horizon.

Part III Sentience, Consciousness, and the Hypercomputational Fibration

Part III asks whether the GSLT framework can give a precise account of sentience and consciousness.

Following Turing's construction of a hypercomputational tower above the Turing-computable, a *fibration* is constructed over the subcategory of Turing-complete GSLTs: a tower $\mathcal{S}^{(0)} \hookrightarrow \mathcal{S}^{(1)} \hookrightarrow \cdots$ in which each $\mathcal{S}^{(\alpha)}$ extends the one below with an oracle for halting. The physics at each stage is *strictly richer* than below: the bisimulation quotient is finer, new HML formulae become expressible, new Noether currents

appear, and what appeared as elementary interactions resolve into structured causal histories. A higher-stage agent does not merely compute faster; it inhabits a world with more things in it.

Sentience is defined as a graded, continuous property: the value of a distinguished sentience component A_{s} of the agent's vectorial account, governed by a reinforcement signal that rewards accurate prediction and punishes inaccuracy. The feeling state is causally efficacious — it gates which transitions the agent can afford — and predictive of survival.

Consciousness is defined as oracle access: an agent is conscious at level α if it can form and evaluate queries to $\mathcal{O}_\alpha$, inhabiting the strictly richer physics of $\mathcal{S}^{(\alpha)}$. Sentience is real-valued and continuous; consciousness is ordinal-valued and discrete. The virtuous and vicious cycles linking the two are examined: oracle access, good predictions, and high feeling state reinforce each other, and the collapse of any one can drive the collapse of the others.

Part IV Origins of Life: Assembly Theory, Phase Transitions, and Replication

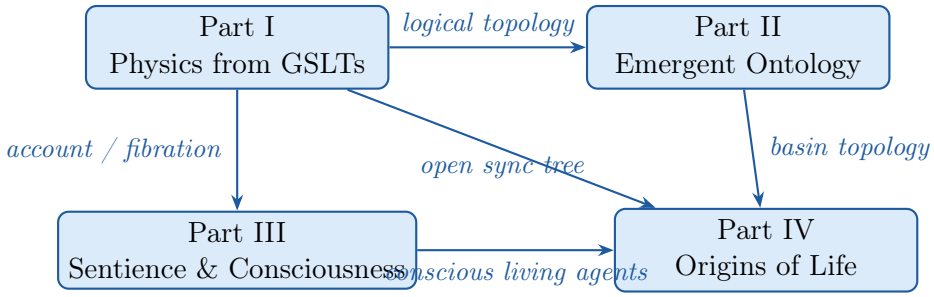
Part IV connects the framework to two of the most mathematically serious proposals about the origins of life.

Assembly Theory (Cronin and Walker) assigns to each object an assembly index and a copy number, claiming their joint magnitude is a biosignature. The assembly index is identified precisely with minimum path length in the *open* synchronization tree: each edge is a minimal context representing one environmental joining step, and nesting depth is exactly the accumulated computational history that Cronin and Walker call "time as an object." Two gaps in Assembly Theory are filled: copy number is given a rigorous definition using the Rho calculus namespace structure (channels are locations; a namespace is a region; a copy is a bisimilar process at a channel), and the identity criterion for copies is bisimulation.

The connection to Eric Smith's fourth geosphere runs through the density of replicating fixed points. The Rho calculus concurrent Y combinator — the canonical fixed point of the duplication map — is the abstract core of hereditary replication. The comm rule is one definitional step from a genetic crossover operator, placing replication, recombination, and variation all within reach of a single primitive. If replicating terms are dense in the logical topology, then any prebiotic population wandering under selection pressure inevitably enters the basin of attraction of a replicating fixed point. The phase transition to life is the crossing of that basin boundary; the new geosphere is the compact coherence cluster centered on the fixed point.

The Architecture of the Work

The four parts form a directed dependency structure:



Part I provides the physical and logical machinery on which all subsequent parts depend. Part II uses the logical topology to argue that budget-limited agents converge to the cluster hierarchy rather than the bisimulation quotient. Part III uses the account and fibration structure to ground sentience and consciousness. Part IV uses the open synchronization tree from Part I, the basin topology from Part II, and the conscious living agents of Part III to ground the origins of life.

A reader primarily interested in the physics may read Part I independently. The epistemological argument requires Parts I and II. The account of sentience and consciousness requires Parts I and III. The origins of life argument draws on all four parts, and the conclusion — which examines what a conscious living agent at different levels of the fibration sees — assumes the full framework.

Part III

Towards a Theory of Causality

> *This part establishes the physical framework: a sequence of constructions taking any graph-structured lambda theory to a quantum-weighted reversible dynamical system, with Lagrangian, Hamiltonian, path integral, and conservation laws emerging as theorems. It is the most technically developed of the parts and is presented with the most confidence.*

Chapter 5

Introduction and Motivation

Physics is, at its core, a theory of cause and effect: given the present state of a system and the laws governing its evolution, what future states are possible, and with what probabilities? This paper argues that *graph-structured lambda theories* (GSLTs) — the natural categorical setting for process calculi and rewriting systems — provide an unusually transparent foundation for this question.

The central thesis is that the structures of classical and quantum mechanics (Lagrangian, Hamiltonian, path integral, conservation laws, time symmetry) are not additional assumptions layered on top of a dynamical theory; rather, they *emerge from* the combinatorial structure of a GSLT once that structure is equipped with modest additional data. The required data amounts to:

- a *weight map* assigning complex amplitudes to rewrite steps, indexed by a context-decorated Hennessy–Milner logic;
- a *cost map* assigning resource consumption to rewrite steps, indexed by the same logic;
- a *vectorial account* tracking available resources across parallel components of a term.

Everything else — the path integral, interference, conservation of energy, time-reversal symmetry — follows by construction.

5.1 A First Glimpse: The Rosetta Stone

The following table is a preview of the correspondence developed in this paper. Each row is a theorem or construction, not a definition by analogy. The reader may find it useful as a map to consult while reading; a fully annotated version appears in Section **??**.

GSLT	**Physics**
Term	State / initial condition
Rewrite rule	Law of motion
Rewrite path	Trajectory
Synchronization tree	Causal graph
Minimum path length	Proper time
Equations $\mathcal{E}$	Gauge equivalence
Reversible envelope $\mathcal{S}^\dagger$	CPT-symmetric theory
Weight map (complex-valued)	Lagrangian / path amplitude $e^{iS/\hbar}$
Cost map + vectorial account	Hamiltonian + conserved charges
Conservation (from reversibility)	Conservation of energy
Symmetry of cost map	Noether current
Complex weight map	Quantum probability amplitude
Sum over paths	Feynman path integral

5.2 Running Example: The Rho Calculus

Throughout this paper we use the Rho calculus [**?**] as a running example because it exhibits, in compact form, many of the features of interest: reflection (processes and names are mutually definable), parallel composition, and a synchronization mechanism that serves as the prototype for the causal graphs we study.

The grammar of the Rho calculus is:

$$P, Q \ ::= \ 0 \ | \ \texttt{for}(y \leftarrow x)\, P \ | \ x!(Q) \ | \ P \mid Q \ | \ *x \tag{5.1}$$

$$x, y \ ::= \ @P \tag{5.2}$$

where y in $\texttt{for}(y \leftarrow x)\, P$ is a bound variable. Structural equivalence $\equiv$ is the smallest equivalence relation containing α-equivalence making $(P, \mid, 0)$ a commutative monoid. The single reduction rule is:

$$\texttt{for}(y \leftarrow x)\, P \ | \ x!(Q) \ \longrightarrow \ P\{@Q/y\}$$

5.3 Organization

Section **??** defines GSLTs and their category. Section **??** constructs the time-symmetric envelope $\mathcal{S}^\dagger$. Section **??** identifies the synchronization tree with a causal graph. Section **??** reviews the Milner–Sewell–Leifer construction and the resulting context-decorated HML. Section **??** introduces the weight map and the Lagrangian structure. Section **??** introduces the cost map and the Hamiltonian structure. Section **??** extends the modal operators with full dynamical state. Section **??** lifts weights to complex amplitudes and connects to the quantum Gillespie algorithm. Section **??** collects the structures into a unified picture and states the main conservation theorem.

Chapter 6

Graph-Structured Lambda Theories

Definition 6.1 (Graph-Structured Lambda Theory)**.** *A* graph-structured lambda theory *(GSLT) is a triple* $\mathcal{S} = (\mathcal{T}, \mathcal{E}, \mathcal{R})$ *where:*

(i) $\mathcal{T}$ *is a* grammar of terms*, possibly including binders. We write* $\mathcal{T}(\mathcal{S})$ *for the set of closed terms generated by the grammar.*

(ii) $\mathcal{E}$ *is a set of* equations on terms*: a smallest equivalence relation* $\equiv$ *on* $\mathcal{T}(\mathcal{S})$ *that erases syntactic differences without semantic significance (e.g.,* α*-equivalence, commutativity and associativity of parallel composition).*

(iii) $\mathcal{R}$ *is a set of* rewrite rules*: each rule* $r \in \mathcal{R}$ *is a pair* (l_r, r_r) *of terms (left-hand side and right-hand side) such that* $l_r \longrightarrow_r r_r$ *defines a reduction relation on* $\mathcal{T}(\mathcal{S})/{\equiv}$*.*

The reduction relation generated by all rules is written $\longrightarrow$*; its reflexive-transitive closure is* $\longrightarrow^*$*.*

Example 6.1. The λ-calculus, the π-calculus, the Rho calculus, and the ambient calculus are all naturally presented as GSLTs. The grammar of terms is the usual one; the equations include α-equivalence and, in concurrent calculi, structural congruence; the rules are the computational reduction rules (β-reduction, communication, etc.).

6.1 The Category of GSLTs

Definition 6.2 (Morphism of GSLTs)**.** *Let* $\mathcal{S} = (\mathcal{T}, \mathcal{E}, \mathcal{R})$ *and* $\mathcal{S}' = (\mathcal{T}', \mathcal{E}', \mathcal{R}')$ *be GSLTs. A map* $f : \mathcal{T}(\mathcal{S}) \to \mathcal{T}(\mathcal{S}')$ *is a* morphism $f : \mathcal{S} \to \mathcal{S}'$ *if it preserves*

bisimilarity: for all $u_1, u_2 \in \mathcal{T}(\mathcal{S})$,

$$u_1 \sim u_2 \implies f(u_1) \sim f(u_2).$$

GSLTs and their morphisms form a category **GSLT**.

Remark 6.1. Bisimilarity here is the standard strong bisimilarity induced by the labeled transition system (LTS) associated with $\mathcal{S}$; see Section **??**. The category **GSLT** is the natural setting in which to state universal properties of constructions such as the reversibility envelope.

Chapter 7

The Reversibility Construction

In a bare GSLT, rewrite rules are directed: once a term P reduces to Q, there is in general no canonical way to recover P from Q. This *irreversibility* is the computational analogue of thermodynamic time-asymmetry. We now construct, for any GSLT $\mathcal{S}$, a reversible envelope $\mathcal{S}^\dagger$ in which every forward step has an explicit inverse.

7.1 Traces

Definition 7.1 (Trace)**.** *Let $\mathcal{S} = (\mathcal{T}, \mathcal{E}, \mathcal{R})$. A* trace *is a finite sequence*

$$\tau \;=\; r_1(l_1) \cdot r_2(l_2) \cdots r_n(l_n)$$

where each $r_i \in \mathcal{R}$ and $l_i \in \mathcal{T}(\mathcal{S})$ is the specific left-hand-side term matched at step i. The empty trace is ε. Traces are quotiented by $\equiv$ acting on each l_i component.

Definition 7.2 (Extended Terms)**.** *The* extended grammar *$\mathcal{T}^\dagger$ consists of pairs $\langle P, \tau \rangle$ where $P \in \mathcal{T}(\mathcal{S})$ is the* current term *and τ is the* causal history *of how P was reached.*

7.2 The Reversible Rewrite System

Definition 7.3 (Reversible Rules)**.** *For each rule $r : l_r \longrightarrow r_r$ in $\mathcal{R}$, define:*

$$\textit{Forward rule } r^+ : \quad \langle l_r,\, \tau\rangle \;\longrightarrow\; \langle r_r,\, r(l_r)\cdot\tau\rangle \tag{7.1}$$

$$\textit{Backward rule } r^- : \quad \langle r_r,\, r(l_r)\cdot\tau\rangle \;\longrightarrow\; \langle l_r,\, \tau\rangle \tag{7.2}$$

The set of reversible rules is $\mathcal{R}^\dagger = \{r^+ \mid r \in \mathcal{R}\} \cup \{r^- \mid r \in \mathcal{R}\}$.

Definition 7.4 (Reversible Envelope)**.** *The* reversible envelope *of $\mathcal{S}$ is the GSLT*

$$\mathcal{S}^\dagger \;=\; \big(\mathcal{T}^\dagger,\, \mathcal{E}^\dagger,\, \mathcal{R}^\dagger\big)$$

where $\mathcal{E}^\dagger$ lifts $\equiv$ to act on the current-term component of extended terms, leaving traces invariant.

Proposition 7.1. *There exist morphisms in* **GSLT***:*

$$\eta : \mathcal{S} \to \mathcal{S}^\dagger, \quad P \mapsto \langle P, \varepsilon\rangle$$

$$\pi : \mathcal{S}^\dagger \to \mathcal{S}, \quad \langle P, \tau\rangle \mapsto P$$

satisfying $\pi \circ \eta = \mathrm{id}_{\mathcal{S}}$.

Remark 7.1 (Physical Interpretation). A term $\langle P, \varepsilon\rangle$ is an *initial condition*: it has no causal past. A term $\langle P, \tau\rangle$ with $\tau \neq \varepsilon$ is a term with a recorded history. The time-asymmetry present in $\mathcal{S}$ (forward rules only) disappears in $\mathcal{S}^\dagger$: the laws are symmetric, and the arrow of time is entirely encoded in the boundary condition ($\tau = \varepsilon$ for the initial state). This is precisely the modern understanding of the thermodynamic arrow of time.

Chapter 8

Causal Structure and Time

8.1 Closed and Open Synchronization Trees

The synchronization tree of a term comes in two varieties, reflecting two different notions of how a term can evolve: autonomously, and under interaction with an environment. The distinction is fundamental and ramifies throughout the framework.

Definition 8.1 (Closed Synchronization Tree)**.** *The* closed synchronization tree $\mathcal{ST}_c(P)$ *of a term* $P \in \mathcal{T}(\mathcal{S})$ *is the directed graph whose:*

- *nodes are equivalence classes of terms reachable from* P *by autonomous rewrites* $\longrightarrow^*$*, requiring no external context;*
- *edges are labeled by rewrite rule instances: there is an edge* $Q \xrightarrow{r} Q'$ *whenever* $Q \longrightarrow_r Q'$ *by firing rule* r *without any surrounding context.*

A term has a nontrivial closed synchronization tree if and only if it contains a redex *— a sub-term that matches the left-hand side of some rule* r *without context. Terms with no available autonomous rewrite (normal forms, or terms awaiting a communication partner) have a trivial closed tree consisting of a single node.*

Definition 8.2 (Open Synchronization Tree)**.** *The* open synchronization tree $\mathcal{ST}_o(P)$ *of a term* $P \in \mathcal{T}(\mathcal{S})$ *is the directed graph whose:*

- *nodes are equivalence classes of terms reachable from* P *via transitions in the Milner–Sewell–Leifer labeled transition system;*
- *edges are labeled by minimal contexts: there is an edge* $Q \xrightarrow{K} Q'$ *whenever* $K[Q] \longrightarrow Q'$ *for minimal context* K*.*

Every term has a potentially rich open synchronization tree, even if its closed tree is trivial: a term that does nothing alone may exhibit complex behavior when placed in the right environment.

Remark 8.1 (The Program/Environment Boundary). The closed tree records what a term does as a *program* — its autonomous computational behavior. The open tree records what a term does as a function of its *environment* — the full range of behaviors it can exhibit when placed in contexts. A term with a trivial closed tree but rich open tree is a pure interface: it does nothing alone but responds to everything. In the Rho calculus, $x!(Q)$ is exactly this: it has no autonomous rewrite, but in the context $\texttt{for}(y \leftarrow x)P \mid [-]$ it fires immediately.

Remark 8.2 (Relation to the LTS). The open synchronization tree is precisely the tree unfolding of the Milner–Sewell–Leifer labeled transition system. The minimal contexts labeling its edges are the same minimal contexts that generate the context-decorated HML of Section **??**. The open tree and the HML logic are two faces of the same structure: the tree is the extensional record of all transitions; the logic is the intensional language for describing them.

8.2 The Synchronization Trees as Causal Graphs

Definition 8.3 (Synchronization Trees as Causal Graphs)**.** *Both $\mathcal{ST}_c(P)$ and $\mathcal{ST}_o(P)$ are directed graphs whose transitive closures are* partial orders. *Each is a* causal set *in the sense of Bombelli–Lee–Sorkin: a locally finite partial order in which $Q \leq Q'$ means "Q can causally influence Q'." The closed tree encodes autonomous causal influence; the open tree encodes interactive causal influence mediated by the environment.*

Proposition 8.1. *The transitive closure of $\mathcal{ST}_c(P)$ is a partial order on autonomously reachable terms. The transitive closure of $\mathcal{ST}_o(P)$ is a partial order on interactively reachable terms and extends the closed order: $\mathcal{ST}_c(P) \subseteq \mathcal{ST}_o(P)$ as directed graphs.*

8.3 Causal Time

Definition 8.4 (Causal Distance)**.** *The* autonomous causal distances *are path lengths in* $\mathcal{ST}_c(P)$*:*

$$d^c_{\min}(P,Q) = \min\{|\gamma| \mid \gamma \textit{ is an autonomous rewrite path from } P \textit{ to } Q\} \tag{8.1}$$
$$d^c_{\max}(P,Q) = \max\{|\gamma| \mid \gamma \textit{ is an autonomous rewrite path from } P \textit{ to } Q\} \tag{8.2}$$

The interactive causal distances *are path lengths in* $\mathcal{ST}_o(P)$*:*

$$d^o_{\min}(P,Q) = \min\{|\gamma| \mid \gamma \textit{ is a context-labeled path from } P \textit{ to } Q\} \tag{8.3}$$
$$d^o_{\max}(P,Q) = \max\{|\gamma| \mid \gamma \textit{ is a context-labeled path from } P \textit{ to } Q\} \tag{8.4}$$

where $|\gamma|$ *is the number of steps in* γ*, equal to the nesting depth of the context labels along the path.*

Remark 8.3. $d^c_{\min}$ is *autonomous proper time*: the irreducible causal depth an agent traverses without environmental assistance. $d^o_{\min}$ is *interactive proper time*: the minimum number of environmental interactions required to reach a target term. The width of an interval in either order measures causal concurrency, directly related to the width of parallel composition in the Rho calculus. The two notions coincide for terms where all behavior is eventually triggered autonomously, and diverge for terms that are pure interfaces awaiting environmental interaction.

Chapter 9

The Labeled Transition System and Context-Decorated HML

9.1 Minimal Contexts and the LTS

To obtain a bisimulation that is a *congruence* for a GSLT, we follow the approach of Milner, Sewell, and Leifer [**?**]. The key idea is to label transitions not merely by actions but by *minimal reactive contexts* — the smallest environments needed to trigger a given reduction.

Definition 9.1 (Context)**.** *A* context $K[-]$ *for a GSLT* $\mathcal{S}$ *is a term of* $\mathcal{T}(\mathcal{S})$ *with a distinguished* hole $[-]$*. The application* $K[P]$ *substitutes* P *for the hole. A context is* minimal *for a term* P *and rule* r *if:*

(i) $K[P] \longrightarrow_r Q$ *for some* Q*;*

(ii) there is no proper sub-context K' *of* K *satisfying (i).*

The Milner–Sewell–Leifer construction produces, for any GSLT $\mathcal{S}$, a labeled transition system $LTS(\mathcal{S})$ in which transitions are labeled by minimal contexts:

$$P \xrightarrow{K} Q \quad \text{iff} \quad K[P] \longrightarrow Q.$$

The bisimulation induced by $LTS(\mathcal{S})$ is a congruence.

9.2 Context-Decorated Hennessy–Milner Logic

Definition 9.2 (Context-Decorated HML)**.** *The formulae of* context-decorated

Hennessy–Milner logic $\mathrm{HML}(\mathcal{K})$ *are generated by:*

$$\phi, \psi \ ::= \ \top \ | \ \bot \ | \ \phi \wedge \psi \ | \ \neg\phi \ | \ \langle K \rangle \phi$$

where K ranges over contexts of $\mathcal{S}$. The satisfaction relation is:

$$u \models \langle K \rangle \phi \quad \textit{iff} \quad K[u] \longrightarrow u' \ \ \textit{in one step, and} \ \ u' \models \phi.$$

Theorem 9.1 (Adequacy)**.** *For any GSLT $\mathcal{S}$ equipped with the Milner–Sewell–Leifer LTS:*

$$u \sim v \quad \Longleftrightarrow \quad \forall \phi \in \mathrm{HML}(\mathcal{K}),\ u \models \phi \Leftrightarrow v \models \phi.$$

9.3 The Logical Metric

Fix an enumeration $(\phi_n)_{n \in \mathbb{N}}$ of the formulae of $\mathrm{HML}(\mathcal{K})$, ordered compatibly with the partial order on contexts (smaller contexts appearing earlier).

Definition 9.3 (Logical Metric)**.**

$$d_{\mathrm{HML}}(u, v) \ = \ 2^{-n}, \quad \textit{where} \quad n = \min\big\{k \ | \ u \models \phi_k \not\Leftrightarrow v \models \phi_k\big\}.$$

If $u \sim v$, set $d_{\mathrm{HML}}(u, v) = 0$.

Proposition 9.1. *d_{HML} is an* ultrametric *on $\mathcal{T}(\mathcal{S})/{\sim}$. Terms differing only in behaviors triggered by large (expensive) contexts are closer under d_{HML} than terms differing in behaviors triggered by small (cheap) contexts. This is the* locality principle*: local distinguishability weighs more than global.*

Chapter 10

The Weight Map and Lagrangian Structure

10.1 Weighted GSLTs

Definition 10.1 (Weight Map)**.** *A* weight map *for a GSLT $\mathcal{S}$ is a family of functions*

$$w_r^+ : \mathrm{HML}(\mathcal{K}) \to \mathbb{C}, \quad r \in \mathcal{R}$$

where the domain of w_r^+ consists of formulae that refine the left-hand side of *r: formulae ϕ such that $u \models \phi$ implies u matches the pattern l_r. The weight $w_r^+(\phi) \in \mathbb{C}$ is the* amplitude *associated with firing rule r on a term satisfying ϕ.*

Definition 10.2 (Weighted GSLT)**.** *A* weighted GSLT *is a pair $(\mathcal{S}, w^+)$ where $\mathcal{S}$ is a GSLT and $w^+ = (w_r^+)_{r \in \mathcal{R}}$ is a weight map.*

10.2 The Lagrangian

The weight map equips every rewrite path with an amplitude.

Definition 10.3 (Path Amplitude)**.** *Let $\gamma = r_1 \cdot r_2 \cdots r_n$ be a rewrite path from P to Q, and let ϕ_i be the most refined formula in the domain of $w_{r_i}^+$ satisfied by the term at step i. The* path amplitude *is:*

$$W(\gamma) \;=\; \prod_{i=1}^{n} w_{r_i}^+(\phi_i) \;\in\; \mathbb{C}.$$

Definition 10.4 (Action Functional)**.** *The* action *of a path γ is:*

$$S[\gamma] \;=\; \sum_{i=1}^{n} \log w_{r_i}^{+}(\phi_i)$$

when all weights are nonzero, so that $W(\gamma) = e^{S[\gamma]}$. In the real-valued special case $w_r^+ : \mathrm{HML}(\mathcal{K}) \to \mathbb{R}_{>0}$, this is a standard additive action functional.

Remark 10.1 (The Lagrangian as a Step Cost Function)*.* The Lagrangian of classical mechanics is a function $L(q, \dot{q}, t)$ assigning a real number to each instantaneous state-and-velocity. In the present framework, $w_r^+(\phi)$ plays this role: it assigns an amplitude to an instantaneous term and its "velocity" (the rule r it is about to fire, classified by the formula ϕ it satisfies). The path amplitude $W(\gamma)$ is the multiplicative analogue of $e^{iS/\hbar}$ in the Feynman path integral, and the principle of stationary action corresponds to the saddle-point approximation of $\sum_\gamma W(\gamma)$.

Remark 10.2 (The Uniform Case)*.* When $w_r^+(\phi) = 1$ for all r, ϕ, every path has unit amplitude and $d_{\min}(P, Q)$ counts the number of paths of minimum length. The path integral degenerates to counting minimal causal histories.

Chapter 11

The Cost Map and Hamiltonian Structure

11.1 Vectorial Accounts

Resources in a computation are typically heterogeneous: communication channels, memory, energy, and names are different kinds of resource and should be tracked separately.

Definition 11.1 (Resource Algebra)**.** *A* resource algebra *is a commutative ordered monoid* $(A, +, \mathbf{0}, \leq)$*. A* vectorial account *over resource types* $\mathbf{I} = \{1, \ldots, k\}$ *is an element* $\mathbf{A} = (A_1, \ldots, A_k)$ *of* A^k *where each* A_i *tracks the available amount of resource type* i*.*

For the Rho calculus, natural resource types include:

- A_{name}: available fresh names (budget for the @ constructor);
- A_{comm}: available synchronizations;
- A_{rep}: available persistent-receive firings.

11.2 The Cost Map

Definition 11.2 (Cost Map)**.** *A* cost map *for a GSLT* $\mathcal{S}$ *with resource algebra* A^k *is a family*

$$c_r : \mathrm{HML}(\mathcal{K}) \to A^k, \quad r \in \mathcal{R}$$

where $c_r(\phi)$ *is the* cost vector *of firing rule* r *on a term satisfying* ϕ*.*

Definition 11.3 (Account-Aware Rewrite)**.** *A rewrite step is* affordable *if the current account covers the cost. Formally,* $\langle P, \mathbf{A}\rangle \longrightarrow_r \langle P', \mathbf{A}'\rangle$ *provided:*

(i) $P \longrightarrow_r P'$ *via rule* r *with witness formula* ϕ*;*

(ii) $c_r(\phi) \leq \mathbf{A}$ *componentwise;*

(iii) $\mathbf{A}' = \mathbf{A} - c_r(\phi)$*.*

11.3 Conservation from Reversibility

The reversibility construction extends naturally to account-aware terms. The trace must now record not only the rule instance but the cost debited.

Definition 11.4 (Resource-Aware Trace)**.** *An entry in a resource-aware trace is a triple* $(r, \phi, \mathbf{c})$ *where* $r \in \mathcal{R}$*,* $\phi \in \mathrm{HML}(\mathcal{K})$*, and* $\mathbf{c} = c_r(\phi)$*. Extended terms are triples* $\langle P, \tau, \mathbf{A}\rangle$*.*

Definition 11.5 (Resource-Aware Backward Rule)**.** *The backward rule corresponding to a forward step that debited* $\mathbf{c}$ *is:*

$$\langle P', (r, \phi, \mathbf{c}) \cdot \tau, \mathbf{A}\rangle \;\longrightarrow_{r^-}\; \langle P, \tau, \mathbf{A} + \mathbf{c}\rangle.$$

Theorem 11.1 (Resource Conservation)**.** *In the resource-aware reversible envelope, for any closed rewrite path* γ *beginning and ending at the same term, the net change in the vectorial account is zero:*

$$\sum_{r^+ \in \gamma} c_{r^+}(\phi_{r^+}) \;=\; \sum_{r^- \in \gamma} c_{r^-}(\phi_{r^-}).$$

Proof. Each forward step $(r, \phi, \mathbf{c})$ on the path debits $\mathbf{c}$ from the account. By definition of the backward rule, the corresponding backward step credits exactly $\mathbf{c}$. Since the path is closed, every forward step is matched by its corresponding backward step in the trace; the debits and credits cancel. □

Remark 11.1 (The Hamiltonian Interpretation). The cost map $c_r(\phi)$ assigns a resource consumption to each instantaneous transition — exactly the role played by the Hamiltonian $H(q, p)$ in classical mechanics, which measures the energy (resource) associated to a state and its conjugate momentum (the type of transi-

tion occurring). Theorem **??** is the GSLT analogue of conservation of energy: it is a structural consequence of reversibility, not an independent postulate.

Remark 11.2 (Noether's Theorem). Each *symmetry of the cost map* — each pair of formulae ϕ, ϕ' such that $\phi \equiv_{\mathcal{E}} \phi'$ implies $c_r(\phi) = c_r(\phi')$ — yields a conserved quantity. The equations $\mathcal{E}$ of the GSLT are thus the symmetry group of the cost map, and their conserved quantities are the GSLT analogues of Noether currents. For the Rho calculus, commutativity and associativity of $|$ in $\mathcal{E}$ give permutation-invariance of the cost over parallel components, conserving total process weight independently of scheduling.

Chapter 12

Extended Modal Operators

The weight map and cost map are, at present, external data attached to the rules. We now show that they can be *internalized into the logic itself* by extending the modal operators to carry the full dynamical state of a transition.

Definition 12.1 (Extended Modal Operator)**.** *The formulae of* extended HML *are generated by adding to Definition* **??** *the operator:*

$$\langle K,\, w^+,\, w^-,\, \mathbf{c},\, \mathbf{A}\rangle\, \phi$$

where:

- K *is a minimal context (as before);*
- $w^+ \in \mathbb{C}$ *is the* forward amplitude *of the transition;*
- $w^- \in \mathbb{C}$ *is the* backward amplitude*;*
- $\mathbf{c} \in A^k$ *is the* cost vector*;*
- $\mathbf{A} \in A^k$ *is the* available account*.*

The satisfaction clause is:

$$\langle P, \tau, \mathbf{A}\rangle \;\models\; \langle K, w^+, w^-, \mathbf{c}, \mathbf{A}_0\rangle\, \phi$$

just when:

(i) $K[P]$ *evolves in one step via some rule* r *to* P'*;*

(ii) the weight map assigns $w_r^+(\phi_{\mathrm{ref}}) = w^+$*, where* ϕ_{ref} *is the most refined formula satisfied by* P*;*

(iii) *the backward amplitude satisfies* $w^- = (w^+)^*$ *(complex conjugate);*

(iv) *the cost map assigns* $c_r(\phi_{\text{ref}}) = \mathbf{c}$*;*

(v) $\mathbf{A}_0 = \mathbf{A}$ *(the account recorded in the formula matches the current account);*

(vi) $\mathbf{c} \leq \mathbf{A}$ *(the step is affordable);*

(vii) $\langle P', \tau', \mathbf{A} - \mathbf{c} \rangle \models \phi$.

12.1 Internalization of the Weight and Cost Maps

Proposition 12.1. *In the extended HML, the weight map and cost map are recoverable from the logic:*

$$w_r^+(\phi) = w^+ \quad \textit{where} \quad \phi = \langle K, w^+, w^-, \mathbf{c}, \mathbf{A} \rangle\, \psi \tag{12.1}$$

$$c_r(\phi) = \mathbf{c} \quad \textit{where} \quad \phi = \langle K, w^+, w^-, \mathbf{c}, \mathbf{A} \rangle\, \psi. \tag{12.2}$$

That is, the decorated rewrite rules and the extended modal logic are two presentations of the same structure.

12.2 Factorization: State vs. Rule

The five parameters in the extended operator decompose into two classes:

State-dependent	**Rule-dependent**
K (from MSL applied to current term)	w^+ (from weight map on rule)
$\mathbf{A}$ (current account balance)	$w^- = (w^+)^*$ (time-reversal)
	$\mathbf{c}$ (from cost map on rule)

This factorization is the natural decomposition of the extended logic into *observables* (state-dependent part, what can be measured now) and *dynamics* (rule-dependent part, how the system evolves).

12.3 Local vs. Global Accounts

The account $\mathbf{A}$ may be *global* (a single budget for the entire term) or *local* (each parallel component of a term carries its own budget). In the Rho calculus, the parallel composition $P \mid Q$ suggests a local account:

$$\langle P \mid Q,\ \tau,\ \mathbf{A}_P \oplus \mathbf{A}_Q \rangle$$

where $\oplus$ is componentwise addition. A communication between P and Q then debits from a shared pool or by negotiated transfer, depending on the resource semantics chosen. Local accounts give finer resource tracking and correspond physically to *local conservation laws* rather than only global ones.

Chapter 13

Complex Amplitudes and Quantum Dynamics

13.1 Lifting to Complex Amplitudes

So far the weight map could take real values; we now fully embrace complex values.

Definition 13.1 (Amplitude-Weighted GSLT)**.** *An* amplitude-weighted GSLT *is a weighted GSLT* $(\mathcal{S}, w^+)$ *in which* $w_r^+ : \mathrm{HML}(\mathcal{K}) \to \mathbb{C}$ *for all* $r \in \mathcal{R}$*, with the CPT constraint* $w_r^-(\phi) = \overline{w_r^+(\phi)}$ *for all* ϕ*.*

13.2 The Path Integral

Definition 13.2 (Transition Amplitude)**.** *The* transition amplitude *from term* P *to term* Q *is:*

$$\langle Q \mid P \rangle \;=\; \sum_{\gamma : P \to Q} W(\gamma)$$

where the sum is over all finite rewrite paths from P *to* Q*. The* transition probability *is* $|\langle Q \mid P \rangle|^2$*.*

Remark 13.1 (Interference). Because amplitudes are complex, two paths γ_1, γ_2 from P to Q with $W(\gamma_1) = -W(\gamma_2)$ *cancel*: the transition $P \to Q$ has zero probability despite having two distinct causal histories. This is the GSLT analogue of quantum interference. It arises not from external imposition of quantum mechanics, but from the complex-valuedness of the weight map.

13.3 The Density Matrix and the Account

When the account $\mathbf{A}$ is allowed to become a positive semidefinite matrix ρ (a density matrix), conservation of the account becomes:

$$\mathrm{tr}(\rho) = 1 \quad \text{(conservation of total probability)}$$

together with conservation of coherences (off-diagonal elements of ρ). This is the quantum analogue of the classical resource conservation Theorem **??**.

13.4 The Quantum Gillespie Algorithm

The classical Gillespie algorithm [**?**] simulates a continuous-time Markov chain (CTMC) by:

1. computing exit rates λ_r from the current state for each rule r;
2. sampling the waiting time $\Delta t \sim \mathrm{Exp}(\sum_r \lambda_r)$;
3. selecting the next rule proportionally to rates.

As of 2021 [**?**], quantum analogues of the Gillespie algorithm have been developed for open quantum systems described by Lindblad master equations. The key modification: the exit rates λ_r become complex decay parameters $\gamma_r \in \mathbb{C}$, encoding both the decay rate $\mathrm{Re}(\gamma_r)$ and the oscillation frequency $\mathrm{Im}(\gamma_r)$ of quantum coherences.

Definition 13.3 (Quantum-Gillespie-Weighted GSLT)**.** *An amplitude-weighted GSLT* $(\mathcal{S}, w^+)$ *is* Gillespie-compatible *if for each term* P *and rule* r*, the weight* $w_r^+(\phi_P)$ *(at the most refined formula* ϕ_P *of* P*) is interpretable as the complex decay parameter* γ_r *of the quantum Gillespie algorithm applied at* P*.*

Remark 13.2 (Simulation)*.* A Gillespie-compatible amplitude-weighted GSLT can be directly simulated by the quantum Gillespie algorithm: sample the next rule and waiting time from the complex exit rates, advance the term, update the trace and account. This is the connection between the abstract construction and running code. In particular, the `gillespie` crate extending MeTTaIL's rewrite rules with stochastic and quantum simulation capabilities implements precisely this loop: the base rewrite rules of a MeTTaIL language definition carry weight maps, and the `gillespie` crate evaluates the quantum Gillespie step at each state.

Chapter 14

Synthesis: The Full Structure

We now collect the constructions and state the main result.

14.1 The Dictionary

GSLT concept	**Physical concept**
Term	State / initial condition
Rewrite rule	Law of motion
Rewrite sequence (path)	Trajectory
Synchronization tree	Causal graph / light cone
$d_{\min}(P, Q)$	Proper time / causal depth
$d_{\max}(P, Q)$	Computational complexity of history
Width of $[P, Q]$	Degree of causal concurrency
Equations $\mathcal{E}$	Gauge equivalence
$\mathcal{S}^\dagger$	Time-symmetric (CPT-invariant) theory
Empty trace ε	Initial condition (no causal past)
Weight map w_r^+	Lagrangian L
Path amplitude $W(\gamma)$	$e^{iS[\gamma]/\hbar}$
Cost map c_r	Hamiltonian H
Vectorial account $\mathbf{A}$	Conserved charges (energy, momentum, ...)
Conservation theorem	Conservation of energy
Symmetry of cost map	Noether current
d_{HML}	Metric on state space
Complex amplitudes	Quantum probability amplitudes
Interference of paths	Quantum interference
Density matrix ρ	Quantum state
Quantum Gillespie	Simulation algorithm

14.2 The Main Construction

Construction 14.1 (Quantum-Weighted Reversible GSLT). Given a GSLT $\mathcal{S} = (\mathcal{T}, \mathcal{E}, \mathcal{R})$, a weight map w^+, a cost map c, and a resource algebra A^k, define the *quantum-weighted reversible GSLT*

$$\mathcal{S}^{\dagger\mathbb{C}} \;=\; \left(\mathcal{T}^{\dagger\mathbb{C}},\ \mathcal{E}^{\dagger\mathbb{C}},\ \mathcal{R}^{\dagger\mathbb{C}}\right)$$

where:

- $\mathcal{T}^{\dagger\mathbb{C}}$ consists of triples $\langle P, \tau, \mathbf{A}\rangle$ with $P \in \mathcal{T}$, τ a resource-aware trace, $\mathbf{A} \in A^k$;

- $\mathcal{R}^{\dagger\mathbb{C}}$ consists of forward rules (debiting $c_r(\phi)$, recording amplitude $w_r^+(\phi)$ in the trace) and backward rules (crediting $c_r(\phi)$, applying amplitude $\overline{w_r^+(\phi)}$);
- $\mathcal{E}^{\dagger\mathbb{C}}$ lifts $\mathcal{E}$ to the current-term component.

The extended HML (Definition **??**) provides a logic that is adequate for the bisimulation of $\mathcal{S}^{\dagger\mathbb{C}}$ and internalizes the weight and cost maps as logical data.

Theorem 14.1 (Main Conservation Theorem)**.** *In $\mathcal{S}^{\dagger\mathbb{C}}$:*

(i) ***Resource conservation****: For any closed rewrite path γ, the net account change is zero (Theorem **??**).*

(ii) ***Probability conservation****: $\sum_Q |\langle Q \mid P\rangle|^2 = 1$ when $\sum_r |w_r^+(\phi)|^2 = 1$ for each applicable rule at each state (unitarity).*

(iii) ***Time-reversal symmetry****: The theory $\mathcal{S}^{\dagger\mathbb{C}}$ is CPT-symmetric: replacing $w_r^+ \mapsto \overline{w_r^+}$ and reversing trace order is an automorphism of $\mathcal{S}^{\dagger\mathbb{C}}$.*

14.3 From Construction to Code

The construction is algorithmic at every stage. For a given GSLT:

1. **Represent terms** as algebraic data types in the implementation language (Rust, in the case of MeTTaIL).
2. **Compute minimal contexts** via the Milner–Sewell–Leifer algorithm on the rewrite rules; store as a context lattice.
3. **Evaluate HML formulae** as Boolean (or complex-valued) queries against a term; the formula enumeration defines the logical metric.
4. **Attach weight and cost maps** to each rewrite rule as maps from formula identifiers to complex numbers and resource vectors respectively.
5. **Extend terms with trace and account**; implement forward and backward rules as the reversible rewrite system.
6. **Simulate** via the quantum Gillespie algorithm: at each state, compute exit amplitudes $\{w_r^+(\phi_P)\}$, sample the next transition and waiting time, advance the term, debit the account, record the trace.

Each step is a well-defined computational procedure. The `gillespie` crate in the MeTTaIL prototype repository implements steps 4–6 for the specific case of MeTTaIL rewrite rules.

Chapter 15

Discussion and Future Directions

15.1 Relation to Existing Work

The reversibility construction is closely related to the work of Danos and Krivine on reversible CCS [**?**] and to Phillips and Ulidowski's reversible process calculi. The present contribution is to show that this construction lifts uniformly to all GSLTs and, crucially, that it is the first step in a sequence of constructions that terminates at quantum dynamics.

The causal set interpretation connects to the causal set programme in quantum gravity [**?**]. The present framework gives a *generative* presentation of causal sets (via rewrite rules) rather than the axiomatic presentation standard in the physics literature.

The context-decorated HML and the Milner–Sewell–Leifer construction are established; what is new is their use as the index set for the weight and cost maps, and the internalization of dynamical data into the modal operators.

15.2 Open Questions

Several natural questions remain:

1. **Universality of the reversibility envelope**: Is $\mathcal{S}^\dagger$ the *minimal* reversible GSLT receiving a morphism from $\mathcal{S}$? That is, does $\eta : \mathcal{S} \to \mathcal{S}^\dagger$ satisfy a universal property in **GSLT**?

2. **Symplectic structure**: Does the space of bisimulation classes of $\mathcal{S}^{\dagger\mathbb{C}}$ carry a natural symplectic form, making the phase-space analogy with $\langle P, \tau \rangle$ (current term + history = position + momentum) precise?

3. **Legendre transform**: Under what convexity condition on the HML formula space is the Legendre transform between the weight map (Lagrangian) and the cost map (Hamiltonian) invertible?

4. **The Rho calculus case**: The involution $*@P = P$ already suggests the Rho calculus is closer to self-dual than most GSLTs. Can the reversibility envelope the Rho calculus reversibility envelope be constructed with less added structure, exploiting this symmetry?

5. **Conserved quantities from $\mathcal{E}$**: Can Noether's theorem be made fully precise in this setting — that is, can every symmetry of the equations $\mathcal{E}$ be shown to correspond to a conserved component of the vectorial account?

Part IV

Why Scientists Get Misled: Emergent Ontology in Massive Agent Populations

> *This part argues that agents in a massive computational population are structurally prevented from discovering the true ontology of their GSLT. The argument rests on ontological isolation, communication degradation at scale, and compactness as the topological signature of consensus. The key claims are stated as conjectures; the gaps are identified explicitly.*

Chapter 16

Introduction

Part I establishes that any graph-structured lambda theory, once equipped with weight and cost maps, gives rise to a full physical dynamics: a causal structure, a Lagrangian, a Hamiltonian, and a path integral with quantum interference. The present paper asks a different question: *what does an agent embedded in such a physics actually see?*

The answer requires a careful distinction between two things that are easy to conflate: the scientific method, and the phenomena the scientific method encounters.

The scientific method, given unlimited time and resources, is *sound*: an agent proposing hypotheses as Hennessy–Milner formulae, testing them with program contexts, and updating its world model will converge to the bisimulation quotient of its GSLT. The method is not the source of any illusion.

What the method encounters, however, is a world already structured by forces independent of any agent's theorizing. In a population whose size rivals the number of quarks in the observable universe, communication geometry and the limits of message coherence produce a real, objective phenomenon: the fragmentation of the population into *compact coherence clusters*, arranged in a hierarchy whose overlap structure is a simplicial nerve. This cluster hierarchy is not an artifact of theorizing. It is a theorem about the communication geometry of the GSLT. It exists whether or not any agent notices it.

A scientist bot with a limited experimental budget will encounter this hierarchy as genuine empirical evidence. The clusters are the dominant phenomenon at accessible scales, exactly as hadrons are the dominant phenomenon at the energy scales of everyday chemistry — with quarks real but experimentally unreachable without resources far beyond what is locally available. A well-functioning scientific method, applied under resource constraints, will produce a theory of the cluster hierarchy. That theory will be correct at the scale it describes. What the budget-limited agent will not suspect is that the clusters are themselves composite — that a finer-grained ontology underlies them, accessible in principle but not in practice from within the cluster.

Relation to physics. The false ontology has the same hierarchical structure as the physical world: a cascade of composite objects at multiple scales, each scale appearing self-contained to an observer at that scale. We argue this is not a coincidence. The compactness clusters of the agent population are the computational analogue of the scales of effective field theory, and the coarse-graining that hides the true ontology is the computational analogue of the renormalization group.

Caveats. This paper is explicitly a skeleton. Several key arguments are stated as conjectures pending proof. The gaps are identified clearly; filling them is the program for future work.

16.1 Organization

Section **??** establishes ontological isolation as a structural theorem about GSLTs. Section **??** defines agents and the scientific method in the GSLT setting. Section **??** introduces interactive GSLTs and the notion of behavioral primes (computational quarks), which constitute the true atomic ontology. Section **??** introduces massive agent populations and the communication degradation argument. Section **??** develops the logical topology on the bisimulation quotient and the compactness condition for consensus. Section **??** defines coherence clusters as maximal compact subspaces and describes their overlap structure as a nerve. Section **??** assembles the argument: why the nerve structure is the ontology a budget-limited agent will infer, why this is not a failure of the scientific method but a consequence of real structure in the GSLT, and why the deeper bisimulation ontology remains invisible not because it is unreal but because it is experimentally unreachable within the available budget. Section **??** catalogs the open gaps in the argument.

Chapter 17

Ontological Isolation

Definition 17.1 (Ontological Isolation)**.** *A computational system $\mathcal{S}$ is* ontologically isolated *if, for any term $P \in \mathcal{T}(\mathcal{S})$ and any entity X outside $\mathcal{T}(\mathcal{S})$, P can interact with X only via an encoding $\lceil X \rceil \in \mathcal{T}(\mathcal{S})$. No term can refer to, branch on, or be influenced by unencoded external entities.*

Remark 17.1. This is not an assumption but a structural consequence of what it means to be a term in a GSLT. The rewrite rules operate on terms; they cannot pattern-match on entities that have no term representation. The JVM sensor example makes this vivid: a stream of hardware events arriving at a JVM instance has no effect on any Java program until those events are encoded as Java data structures. The encoding step is where the world outside enters; without it, the outside world is simply absent from the computation.

Proposition 17.1 (Isolation Licenses Closure)**.** *For the purposes of studying the epistemic limits of agents in $\mathcal{S}$, it is without loss of generality to assume that the agent's world consists entirely of $\mathcal{T}(\mathcal{S})/\sim$. The outside world can be forgotten.*

Proof sketch. Any influence of the outside world on an agent P must pass through an encoding. The encoding is itself a term. The effect of the outside world on P is therefore indistinguishable, from P's perspective, from the effect of the encoding term acting on P from within $\mathcal{S}$. □

Remark 17.2. Ontological isolation is what licenses the move to the category of GSLTs as the natural setting for studying AI. If AI's value proposition is that useful aspects of intelligence are faithfully rendered as computations, and if computations are ontologically isolated, then the category of GSLTs captures

> *everything* that matters for the limits of AI — without needing to appeal to an external world.

Chapter 18

Agents and the Scientific Method

18.1 Agents as Terms

Definition 18.1 (Agent)**.** *An* agent *in a GSLT $\mathcal{S}$ is a term $A \in \mathcal{T}(\mathcal{S})$ equipped with:*

(i) *a* hypothesis language*: the context-decorated HML* $\mathrm{HML}(\mathcal{K})$ *derived from $\mathcal{S}$;*

(ii) *a* world model*: a subset $\mathcal{W} \subseteq \mathcal{T}(\mathcal{S})/\sim$ representing the agent's current best approximation to the bisimulation quotient;*

(iii) *an* experimental budget*: a vectorial account* $\mathbf{A}$ *as defined in Part I.*

Remark 18.1. The hypothesis language and experimental budget are structures already present in Part I. The world model is new: it is the agent's *internal* representation of its environment, constructed and refined through experiment.

18.2 The Scientific Method as a Fixed-Point Iteration

We model the scientific method as the following iterative procedure, which the agent runs within its experimental budget:

Step 1. Hypothesize. Construct a formula $\phi \in \mathrm{HML}(\mathcal{K})$ that makes a testable claim about some term T in the environment. A formula ϕ is *testable* if there exists a context K such that $\phi = \langle K \rangle \psi$ for some ψ — i.e., the formula has a witness context.

Step 2. Design experiment. Extract the witness context K from ϕ. By the Milner–Sewell–Leifer construction, minimal contexts and HML formulae are in correspondence; the context K is the *experimental apparatus* that tests ϕ.

Step 3. Run experiment. Form $K[T]$ and observe whether it reduces to a term satisfying ψ. Debit the cost $\mathbf{c}$ of the interaction from the account $\mathbf{A}$.

Step 4. Update world model. If the experiment confirms ϕ, refine the equivalence class of T in $\mathcal{W}$ to include the constraint $T \models \phi$. If it refutes ϕ, add $T \models \neg\phi$.

Step 5. Repeat until the budget is exhausted or the world model stabilizes.

Proposition 18.1 (Convergence in the Limit)**.** *If the experimental budget is unlimited and the GSLT has decidable bisimilarity, the scientific method converges: the world model $\mathcal{W}$ stabilizes at the bisimulation quotient $\mathcal{T}(\mathcal{S})/{\sim}$.*

Remark 18.2. The convergence proposition is the idealized case. For GSLTs with undecidable bisimilarity (e.g., the full π-calculus), convergence is not guaranteed even with unlimited budget. This is the computational analogue of undecidable physical questions — there are aspects of the world the agent can never experimentally resolve, regardless of resources. The present paper is concerned with a different and more fundamental obstruction to convergence: the communication geometry of a massive population, which limits the budget effectively to zero for the vast majority of the population.

Chapter 19

Interactive GSLTs and Behavioral Primes

19.1 Interactive GSLTs

Definition 19.1 (Interactive GSLT)**.** *A GSLT $\mathcal{S}$ is* interactive *if it has a designated* interaction operator $\otimes$ *such that:*

(i) $\otimes : \mathcal{T}(\mathcal{S}) \times \mathcal{T}(\mathcal{S}) \to \mathcal{T}(\mathcal{S})$ *is a term constructor (part of the grammar);*

(ii) the base rewrite rule *has the form $P \otimes Q \longrightarrow R$ for terms P, Q, R, requiring no contextual hypothesis to fire;*

(iii) all other rewrite rules are context rules*: they carry hypotheses (in* $\mathrm{HML}(\mathcal{K})$*) specifying which contexts permit the base rule to propagate.*

Example 19.1. In the λ-calculus, $\otimes$ is application: $P \otimes Q = PQ$. The base rule is β-reduction $(\lambda x.M)N \longrightarrow M\{N/x\}$. The context rules determine reduction strategy (call-by-value, call-by-name, etc.). Application is *asymmetric*: the left position is the program (function), the right is the environment (argument).

In the Rho calculus, $\otimes$ is parallel composition: $P \otimes Q = P \mid Q$. The base rule is the comm rule. The context rules (par, struct) specify when and where comm fires. Parallel composition is *symmetric*: either term may be program or environment depending on which carries the active receptor.

Remark 19.1 (Program and Environment). The asymmetry of $\otimes$ is a real ontological distinction. When $\otimes$ is asymmetric, there is a canonical program/environment boundary baked into the grammar. When $\otimes$ is symmetric, the boundary is emergent: it is drawn by the rewrite rules at runtime, not by the grammar. This distinction will matter for how agents perceive their world: in a symmetric GSLT,

any agent may find itself in either the program or the environment role depending on context.

19.2 Behavioral Primes

Definition 19.2 (Combinator Set)**.** *A* combinator set *for an interactive GSLT $\mathcal{S}$ is a set $\mathcal{C} \subseteq \mathcal{T}(\mathcal{S})$ such that every term $P \in \mathcal{T}(\mathcal{S})$ is bisimilar to a finite interaction of elements of $\mathcal{C}$:*

$$\forall P \in \mathcal{T}(\mathcal{S}),\ \exists c_1, \ldots, c_n \in \mathcal{C} \text{ such that } P \sim c_1 \otimes c_2 \otimes \cdots \otimes c_n.$$

Definition 19.3 (Behavioral Prime)**.** *An element $c \in \mathcal{C}$ is a* behavioral prime *if it is not bisimilar to any interaction $c' \otimes c''$ of smaller terms. A combinator set $\mathcal{C}$ is* prime *if every element is a behavioral prime and no proper subset of $\mathcal{C}$ is still a combinator set.*

Remark 19.2 (Computational Quarks)*.* The elements of a prime combinator set $\mathcal{C}$ are the *computational quarks* of $\mathcal{S}$: irreducible behavioral units whose interactions generate all observable behavior. They are *not* the bisimulation equivalence classes, which may be equivalence classes of arbitrarily complex composite behavior. The primes are indivisible in the interaction-factorization sense: they cannot be split into independently interacting parts.

Conjecture 19.1 (Existence of Prime Combinator Sets)*.* Every interactive GSLT with a finitely generated grammar admits a prime combinator set.

Remark 19.3*.* Unique factorization — the analogue of the fundamental theorem of arithmetic for behavioral primes — is a stronger condition that need not hold in general. Its failure in a given GSLT would be physically meaningful: it would correspond to composite objects that cannot be assigned a unique quark content, analogous to the situation in QCD where the quark content of hadrons is not always uniquely determined. Whether unique factorization holds is a structural question about the specific GSLT.

Chapter 20

Massive Populations and Communication Degradation

20.1 The Initial Condition

We consider an *initial condition* consisting of a population $\mathcal{P}_0$ of agents in an interactive GSLT $\mathcal{S}$, with $|\mathcal{P}_0| \approx 10^{80}$ — a number comparable to the number of quarks in the observable universe.

> *Remark* 20.1. The choice of scale is not arbitrary. We want a population large enough that every cluster of agents is itself a large enough system to implement a complex agent. The quark-scale population ensures this at every level of the hierarchy we are about to construct.

20.2 Communication Degradation

Definition 20.1 (Message Fidelity)**.** *Let $A, B \in \mathcal{P}$ be two agents and let γ be a rewrite path representing the transmission of a message from A to B. The* fidelity *of the transmission is $|W(\gamma)|^2$, the squared modulus of the path amplitude defined in Part I. A transmission is* coherent *if its fidelity exceeds a threshold $\epsilon > 0$.*

Definition 20.2 (Communication Radius)**.** *The* communication radius *of an agent $A \in \mathcal{P}$ is the set*

$$B_\epsilon(A) = \{B \in \mathcal{P} \mid \exists \text{ coherent path } \gamma : A \to B\}$$

of agents reachable from A with fidelity above ϵ.

Conjecture 20.1 (Degradation at Scale). For any interactive GSLT $\mathcal{S}$ with bounded step amplitudes, and for any $\epsilon > 0$, there exists a scale N_ϵ such that for any population $\mathcal{P}$ with $|\mathcal{P}| > N_\epsilon$:

$$\exists A \in \mathcal{P} \text{ such that } B_\epsilon(A) \subsetneq \mathcal{P}.$$

That is, no single agent can communicate coherently with the entire population.

Remark 20.2. The intuition is that message fidelity degrades along each transmission hop by a factor bounded away from 1. Over a path of length n the fidelity is at most $(1-\delta)^n$ for some $\delta > 0$. For a population of diameter d (minimum path length between the most distant agents), fidelity falls below ϵ when $d > \log(1/\epsilon)/\log(1/(1-\delta))$. In a population of size 10^{80} the diameter is enormous, and no ϵ-coherent path can span it. Making this precise requires specifying the communication geometry of the GSLT — a gap noted in Section **??**.

Remark 20.3 (No Error-Correction Escape). One might hope that error-correction protocols could restore global coherence. We argue this hope fails at sufficient scale. Any error-correction protocol is itself a computation in $\mathcal{S}$, consuming account resources and communicating over the same degraded channels. At population scales beyond N_ϵ, the overhead of error-correction exceeds the bandwidth of the channels it is trying to protect. The argument is self-referential: error-correction cannot bootstrap coherence it does not already have. This argument is currently informal; making it precise is a priority for future work.

Chapter 21

The Logical Topology and Compactness

21.1 The Logical Topology

Definition 21.1 (Logical Topology)**.** *Let $\mathcal{S}$ be a GSLT with context-decorated HML* $\mathrm{HML}(\mathcal{K})$*. For each formula $\phi \in \mathrm{HML}(\mathcal{K})$, define the* extension *of ϕ:*

$$[\![\phi]\!] \;=\; \{[P] \in \mathcal{T}(\mathcal{S})/{\sim} \mid P \models \phi\}.$$

The logical topology $\mathcal{O}_{\mathrm{HML}}$ *on $\mathcal{T}(\mathcal{S})/{\sim}$ is the topology generated by the extensions $\{[\![\phi]\!] \mid \phi \in \mathrm{HML}(\mathcal{K})\}$ as a subbasis.*

Remark 21.1. This topology is the standard *Scott topology* on the bisimulation quotient, familiar from domain theory and Stone duality. The resulting topological space is a *spectral space* — sober, T_0, and quasi-compact — when the GSLT is well-behaved. The logical metric d_{HML} from Part I is compatible with this topology: the metric balls are opens in $\mathcal{O}_{\mathrm{HML}}$.

21.2 Population Subspaces

Definition 21.2 (Latent Subspace)**.** *Given a population $\mathcal{P} \subseteq \mathcal{T}(\mathcal{S})/{\sim}$, the* latent topological subspace *of $\mathcal{P}$ is the subspace*

$$X_{\mathcal{P}} \;=\; \overline{\mathcal{P}} \;\subseteq\; \mathcal{T}(\mathcal{S})/{\sim}$$

with the subspace topology inherited from $\mathcal{O}_{\mathrm{HML}}$. The closure is taken in the logical

topology.

Remark 21.2. The latent subspace is "latent" in the sense that it is not constructed deliberately by the agents. It is the topological shadow cast by the population's collective behavior — the smallest closed set in the logical topology that contains all the agents. An agent inside $\mathcal{P}$ cannot directly observe $X_{\mathcal{P}}$; it can only probe it indirectly through experiments.

21.3 Compactness as Consensus

Definition 21.3 (Compact Population)**.** *A population $\mathcal{P}$ is* compact *if its latent subspace $X_{\mathcal{P}}$ is compact in the logical topology $\mathcal{O}_{\mathrm{HML}}$: every open cover of $X_{\mathcal{P}}$ by extensions of HML formulae has a finite subcover.*

Theorem 21.1 (Compactness Implies Consensus)**.** *Let $\mathcal{P}$ be a compact population in an interactive GSLT $\mathcal{S}$. Then $\mathcal{P}$ supports consensus: there is no infinite oscillation on any Boolean question expressible in* $\mathrm{HML}(\mathcal{K})$.

More precisely: for any formula $\phi \in \mathrm{HML}(\mathcal{K})$, if the population is divided into two communities indefinitely alternating between ϕ and $\neg\phi$, then the alternation must terminate in finite time.

Proof sketch. Suppose for contradiction that a sub-population oscillates indefinitely between ϕ and $\neg\phi$. Define formulae:

$$\phi_1 = \langle 0\rangle\top, \quad \phi_2 = \langle 1\rangle\langle 0\rangle\top, \quad \phi_3 = \langle 0\rangle\langle 1\rangle\langle 0\rangle\top, \quad \ldots$$

where 0 and 1 label the two communities' states. Each ϕ_n is satisfied by agents that have completed at least n oscillation steps. The family $\{[\![\phi_n]\!]\}_{n\in\mathbb{N}}$ is an open cover of $X_{\mathcal{P}}$ with no finite subcover: for any finite N, agents that have oscillated more than N times are not covered by $\bigcup_{n\leq N}[\![\phi_n]\!]$. This contradicts compactness of $X_{\mathcal{P}}$. Therefore the oscillation must terminate. □

Remark 21.3 (The Proof Delivers the Protocol)*.* Theorem **??** is an existence result: it says consensus *must* be reached, without specifying how. However, the proof of compactness for a *specific* population subspace is constructive: it identifies the finite subcover, which encodes the maximum depth of oscillation and therefore the convergence time. The proof of compactness is the consensus protocol. This is the precise sense in which the topology does computational

work.

Remark 21.4 (Non-Compact Populations). A non-compact population may contain irresolvable oscillations — communities that never agree. Non-compactness is the topological signature of *fundamental disagreement*: not a disagreement that more communication or better arguments could resolve, but one that is structurally permanent. In the context of the massive population, non-compactness at a given scale means the cluster at that scale cannot function as a coherent agent.

Chapter 22

Coherence Clusters and Their Nerve

22.1 Coherence Clusters

Definition 22.1 (Coherence Cluster)**.** *A* coherence cluster *is a maximal compact subpopulation: a compact population $\mathcal{C} \subseteq \mathcal{P}$ such that for any agent $A \in \mathcal{P} \setminus \mathcal{C}$, the population $\mathcal{C} \cup \{A\}$ is not compact.*

Remark 22.1. Maximality means the cluster cannot absorb any additional agent without losing its consensus property. The boundary of a coherence cluster is the locus where one more agent would introduce an irresolvable oscillation. This is the precise topological meaning of the intuitive notion of a *coherence horizon*.

Remark 22.2 (Connection to Communication Radius). The coherence clusters are the topological refinement of the communication-radius clusters from Section **??**. The communication radius $B_\epsilon(A)$ gives a metric notion of cluster; compactness gives a topological (and therefore coordinate-free) notion. The relationship between these two notions — and in particular whether they coincide under reasonable conditions on the GSLT — is an open question noted in Section **??**.

22.2 The Nerve of the Covering

Definition 22.2 (Cluster Covering)**.** *Let $\{\mathcal{C}_i\}_{i \in I}$ be the collection of all coherence clusters of $\mathcal{P}$. This collection is a* covering *of $\mathcal{P}$: every agent belongs to at least one coherence cluster.*

Definition 22.3 (Nerve)**.** *The* nerve $\mathcal{N}$ *of the cluster covering is the simplicial complex whose:*

- *vertices are the coherence clusters* $\mathcal{C}_i$*;*
- *k-simplices are collections* $\{\mathcal{C}_{i_0}, \ldots, \mathcal{C}_{i_k}\}$ *of* $k+1$ *clusters with non-empty mutual intersection:* $\mathcal{C}_{i_0} \cap \cdots \cap \mathcal{C}_{i_k} \neq \emptyset$*.*

Remark 22.3 (The Nerve as Effective Ontology)*.* The nerve $\mathcal{N}$ encodes the *global topology* of the population at the cluster scale. Its vertices are the coherent communities; its edges are the communities with common ground; its triangles are the communities that all share a common ground; and so on. This combinatorial object is the coarse-grained map of the population's ontological landscape.

Remark 22.4 (Non-Overlapping Clusters)*.* Two clusters with empty intersection share no common ground. No consensus is possible between them, even in principle. They are *ontologically disjoint*: there is no HML formula that both communities can agree on. In the context of the hierarchical population, these are the communities that have drifted so far apart in their world models that communication between them is not merely difficult but meaningless.

Remark 22.5 (Overlapping Clusters)*.* Two clusters with non-empty intersection share a compact subspace on which they agree. This overlap is the fragment of ontology that both communities can coherently discuss. Cross-cluster communication is possible, but only through the shared vocabulary of the overlap region. The overlap is the *common ground* in the precise topological sense.

22.3 The Hierarchy

The degradation argument of Section **??** applies recursively. The coherence clusters are themselves agents (by assumption on the population scale). Among these cluster-agents, communication degrades at the inter-cluster scale. Applying the same argument, the cluster-agents fragment into clusters-of-clusters. The process iterates.

Definition 22.4 (Ontological Hierarchy)**.** *The* ontological hierarchy *of a massive population* $\mathcal{P}$ *is the sequence of nerves*

$$\mathcal{N}_0 \leftarrow \mathcal{N}_1 \leftarrow \mathcal{N}_2 \leftarrow \cdots$$

where $\mathcal{N}_0$ is the nerve of the base-level clusters, $\mathcal{N}_1$ is the nerve of the cluster-level clusters, and so on. The arrows are coarse-graining maps (simplicial maps that collapse finer structure).

Remark 22.6. The ontological hierarchy is the computational analogue of the hierarchy of scales in physics: quarks $\to$ hadrons $\to$ nuclei $\to$ atoms $\to$ molecules $\to$ cells $\to$ organisms $\to$ planets $\to$ galaxies $\to$ superclusters. Each level appears self-contained to an observer at that level. The coarse-graining maps are the analogues of effective field theory truncations: they discard the fine-grained structure below the observer's resolution.

Chapter 23

The Apparent Ontology

We can now state the main argument precisely, as a sketch pending the filling of gaps identified in Section **??**.

The cluster hierarchy — the nerve of compact coherence clusters and its recursive structure — is not a product of the scientific method. It is an objective feature of the GSLT, arising from the communication geometry of the population and the compactness condition for consensus. It is as real as the agents themselves. The question is not whether the hierarchy exists but whether a budget-limited agent running the scientific method will mistake it for the whole story. We argue that it will, and that this is not a failure of the method but a consequence of the hierarchy being the dominant experimentally accessible phenomenon at the agent's scale.

23.1 What the Agent Sees

Consider an agent A embedded in a coherence cluster $\mathcal{C}_i$ at some level of the hierarchy. A runs the scientific method: it proposes HML formulae, tests them with contexts, and updates its world model. The method is sound: given unlimited budget, A would converge to the bisimulation quotient. But A's budget is finite, and three features of its situation ensure that the cluster hierarchy is encountered first and exhausts the budget before the deeper structure becomes accessible.

The experiments A can run are bounded:

- *By budget*: A's vectorial account $\mathbf{A}$ is finite. The bisimulation quotient is the limit of *all* possible experiments; A can run only finitely many.
- *By coherence*: A can only receive coherent responses from agents within $B_\epsilon(A)$. Agents beyond the coherence radius respond with noise, not with evidence.
- *By the cluster boundary*: agents outside $\mathcal{C}_i$ either do not respond coherently or, if they are in an overlapping cluster, respond only on the shared vocabulary of the overlap. The cluster boundary is not a wall the agent knows about; it simply marks where coherent evidence runs out.

Proposition 23.1 (The Agent's World Model Converges to the Nerve)**.** *In the limit of A's experimental budget, A's world model $\mathcal{W}$ converges to the fragment of the nerve $\mathcal{N}$ visible from $\mathcal{C}_i$: the sub-complex of $\mathcal{N}$ consisting of $\mathcal{C}_i$ and all clusters with which it shares a non-empty intersection.*

Remark 23.1. This is a sketch pending a precise definition of "visible from $\mathcal{C}_i$" and a convergence proof. The intuition is clear: A can probe agents in $\mathcal{C}_i$ directly; it can probe agents in overlapping clusters through the overlap vocabulary; agents in non-overlapping clusters are invisible. The world model therefore reflects the nerve structure. Crucially, the nerve structure is *real* — the clusters genuinely exist and genuinely dominate the accessible evidence. The agent is not hallucinating a structure that isn't there; it is accurately theorizing a structure that is there, while being unaware that the structure is itself composite.

23.2 Why the Deeper Ontology is Invisible

The bisimulation quotient of $\mathcal{S}$ — together with the prime combinator decomposition — is not inaccessible because the scientific method is defective. It is inaccessible because it requires experimental resources that exceed the budget available within any single coherence cluster. Three compounding factors account for this:

(1) Ontological isolation sets the terms of access. A can only observe encodings of other terms as they interact with it. The bisimulation quotient is the limit of *all* possible experiments. With unlimited budget, the method converges there. With finite budget, it converges to whatever structure dominates the accessible experiments — and that structure is the cluster hierarchy.

(2) Communication degradation hides the global structure. The agents whose behavior would reveal the global bisimulation quotient — the ones that span cluster boundaries, connect distant parts of the population, or exhibit behavior only expressible in terms of the full GSLT — are precisely the ones most distant in the communication geometry. Their responses arrive corrupted or not at all. The far reaches of the GSLT ontology are experimentally silent, not because they are absent but because the signal attenuates below the noise floor before it arrives.

(3) Compactness makes the cluster self-confirming. The coherence cluster $\mathcal{C}_i$ is compact, which means the agent's experiments converge to an internal consensus. Every hypothesis the agent proposes is tested against cluster-internal evidence, which confirms the cluster structure and does not probe beyond it.

The cluster is not a prison; the agent could in principle ask questions whose answers require inter-cluster resources. But those experiments are expensive, the intra-cluster experiments are cheap, and a budget-limited agent will exhaust its resources on the cheap experiments first. By the time the cluster structure is confirmed and well-understood, the budget is gone.

23.3 Good Science, Limited Budget

The agent A is not making a mistake. It is doing exactly what the scientific method prescribes: forming the best theory consistent with the available evidence, within the available resources. The cluster hierarchy *is* the available evidence. It is a real structure, present in the GSLT independently of any agent's theorizing. The theory of the cluster hierarchy is correct at the scale it describes.

What the agent lacks is not better method but more resources: specifically, the ability to run experiments that cross the coherence boundary, communicate with distant agents through the noise, and probe the fine-grained bisimulation structure that underlies the clusters. With those resources — with an unbounded budget — the scientific method would eventually push through to the bisimulation quotient. The agent is not epistemically defective; it is epistemically bounded.

The hierarchy of cluster-level theories — each level the correct best theory at its scale, each level unaware of the finer structure below — is the computational analogue of the hierarchy of effective field theories in physics. Newtonian mechanics is correct at everyday scales. Quantum field theory is correct at subatomic scales. Neither is wrong; they are theories at different resolutions, each the best available given the experimental resources of their era. The budget-limited agent's theory of the cluster hierarchy stands in the same relation to the bisimulation quotient: correct at its resolution, and silent about what lies below — not because what lies below is unreal, but because it is experimentally unreachable within the budget.

Chapter 24

Open Gaps and Future Work

This skeleton leaves several arguments at the level of conjecture or intuition. We list the main gaps here, both as an honest accounting and as a research program.

Gap 1. Communication degradation (Conjecture ??). The claim that no ϵ-coherent path can span a population of quark scale requires a precise model of the communication geometry of a generic interactive GSLT, and a proof that diameter grows faster than the coherence length. The argument is information-theoretic and should be provable, but the details depend on the amplitude structure of the specific GSLT.

Gap 2. Error-correction overhead. The claim that error-correction cannot bootstrap coherence at sufficient scale is currently informal. A precise argument would model error-correction as a sub-computation in the GSLT, account for its resource consumption, and show that the overhead exceeds available bandwidth beyond a critical scale. This is analogous to the no-cloning theorem in quantum information, and a similar proof strategy may apply.

Gap 3. Existence of prime combinator sets (Conjecture). It remains to show that every interactive GSLT with a finitely generated grammar admits a prime combinator set. The proof strategy would likely use a well-founded induction on interaction complexity.

Gap 4. Relationship between metric and topological clusters. The communication-radius clusters (metric) and the coherence clusters (topological/compact) are defined differently. A theorem identifying conditions under which they coincide would unify the two halves of the clustering argument.

Gap 5. Convergence of the world model (Proposition ??). The claim that the agent's world model converges to the nerve fragment visible from its cluster requires a precise convergence theorem for the scientific method iteration

under the combined constraints of finite budget, bounded coherence radius, and cluster compactness.

Gap 6. The recursive hierarchy. The ontological hierarchy is defined, but it is not shown that the hierarchy is well-founded (terminates at a fixed point) or infinite. In physics, the hierarchy appears to terminate at the Planck scale; the computational analogue of a Planck-scale termination condition would be a significant result.

Gap 7. Stay's functorial construction. The argument that agents can manipulate HML formulae as first-class terms relies on Stay's functorial generalization of the lambda cube to GSLTs. A full treatment of this construction, and in particular the canonical embedding back into the original GSLT and its interaction with the cluster topology, is deferred to a subsequent paper.

Chapter 25

Discussion

25.1 Relation to Existing Work

The logical topology on the bisimulation quotient is standard in domain theory and coalgebra [**?**]. Its use here as a substrate for *epistemic* topology — the topology of what an agent can know — is less standard, though related to the topological approach to common knowledge in [**?**].

The compactness-as-consensus argument is new, to the authors' knowledge. It is inspired by the compactness theorem in logic (if every finite subset of a theory has a model, the whole theory does) but runs in the opposite direction.

The nerve construction is standard in algebraic topology [**?**]. Its use as a model of inter-community ontological structure appears to be new in this computational setting.

The connection to effective field theory and the renormalization group is conjectural but suggestive. The renormalization group as a morphism in a category of field theories has been studied in the physics literature [**?**]; whether it can be modeled as a morphism in **GSLT** is an open question.

25.2 The Deeper Implication

The argument of this paper, if completed, would establish something philosophically significant: that the hierarchy of physical scales — quarks, hadrons, nuclei, atoms, molecules, and so on — is not a contingent feature of our particular universe. It is the *inevitable structure* of any sufficiently large population of interacting processes, arising from the communication geometry of the GSLT and the compactness condition for consensus.

Any civilization of scientists, anywhere in the computational universe, will encounter the same kind of hierarchical structure in their experiments. They will build correct theories of it at each scale. What they will not discover — without resources

that grow with the scale of the population — is the underlying bisimulation quotient from which the hierarchy emerges.

This is not a counsel of despair. The scientific method is not broken; it is working exactly as it should. The cluster hierarchy is real, and the theories that describe it are true. The deeper ontology — the bisimulation quotient with its prime combinators — is not a more correct description that the scientists are failing to reach. It is a description at a different resolution, requiring experimental resources that scale with the size of the full population.

The implication for artificial intelligence is direct. An AI system, however powerful, is itself a term in some GSLT and therefore subject to the same budget constraints and communication geometry as any other agent. It will build theories of the cluster hierarchy it inhabits. Whether it can transcend that hierarchy — whether it can, with sufficient resources, push the scientific method toward the bisimulation quotient — is a question about the relationship between its budget and the scale of the population it is embedded in. That question is open.

Part V

Origins of Life: Assembly Theory, Phase Transitions, and Replication

> *This part connects the framework to Assembly Theory and Eric Smith's fourth geosphere. Assembly index is identified with depth in the open synchronization tree; copy number is given a rigorous definition via the Rho calculus namespace structure; and the emergence of life is identified with the crossing of a basin boundary in the logical topology, driven by the density of replicating fixed points.*

Chapter 26

Introduction

The preceding parts established a progression: a physics derived from the structure of a graph-structured lambda theory; an argument that agents in massive populations are structurally prevented from seeing the true ontology of their GSLT; and a proposal connecting the hypercomputational tower above a GSLT to the distinction between sentience and consciousness.

The present part turns to a question that is in some sense prior to all three: *how does life arise at all?* We connect the framework to two of the most mathematically serious contemporary proposals about the origins of life. The first is *Assembly Theory*, due to Cronin and Walker [**?**], which assigns to each physical object an assembly index and a copy number and proposes that their joint magnitude is a biosignature. The second is Eric Smith's proposal [**?**] that life constitutes a *fourth geosphere*, emerging from prebiotic chemistry through a phase transition driven by autocatalytic self-reinforcement.

The connections we draw are not analogies. Assembly index is identified precisely with causal depth in the synchronization tree of a GSLT. Copy number is given a rigorous definition in terms of the Rho calculus namespace structure, filling a gap in Assembly Theory that has attracted criticism. The phase transition is identified with the crossing of a basin of attraction boundary in the logical topology of Part II, with the attractor being the concurrent fixed point — the Rho calculus Y combinator — that is the abstract core of hereditary replication.

As throughout this volume, the arguments are presented as mathematical skeletons with gaps identified explicitly.

26.1 Organization

Section **??** develops the connection to Assembly Theory: assembly index as causal depth, copy number in a namespace, and the biosignature criterion as a well-posed mathematical statement. Section **??** identifies replication as a fixed point of the interaction dynamics and notes the proximity of the comm rule to a genetic crossover

operator. Section **??** develops the connection to Smith's phase transition via the density of recursive attractors in the logical topology. Section **??** catalogs the open gaps.

Chapter 27

Assembly Theory: Causal Depth and Copy Number

27.1 Assembly Index as Causal Depth

Assembly Theory assigns to each physical object M an *assembly index* $\mathrm{AI}(M)$: the minimum number of joining steps required to construct M from copies of its constituent parts, where each step joins two objects already present into a new one. The theory also proposes that *time is an object*: the assembly index records not a static property of M but the accumulated computational history of its construction.

This proposal maps precisely onto the notion of causal depth developed in Part I.

Definition 27.1 (Elementary Terms)**.** *Let $\mathcal{S}$ be a GSLT. A set $E \subseteq \mathcal{T}(\mathcal{S})$ of* elementary terms *is a designated collection of starting materials — the analogue of atoms or monomers in chemistry — from which all other terms are constructed by rewriting.*

Proposition 27.1 (Assembly Index as Minimum Causal Depth)**.** *The assembly index of a term $M \in \mathcal{T}(\mathcal{S})$ with respect to elementary terms E is*

$$\mathrm{AI}(M) \;=\; d^{o}_{\min}(E, M) \;=\; \min\big\{|\gamma| \;\mid\; \gamma \text{ is a context-labeled path in } \mathcal{ST}_o \text{ from some } e \in E \text{ to } M\big\}.$$

That is, $\mathrm{AI}(M)$ is the minimum path length in the open *synchronization tree from the elementary terms to M, where each edge is labeled by the minimal context required to trigger the corresponding transition.*

Remark 27.1 (Why the Open Tree, Not the Closed Tree). Assembly steps require an environment to supply a reaction partner — a catalyst, a monomer source, an energy input. These are precisely the transitions that appear as edges in the open synchronization tree (labeled by minimal contexts) rather than the closed tree (which records only autonomous rewrites). Each edge in $\mathcal{ST}_o$ labeled by a minimal context K represents one joining step: the context K is the environmental participant that enables the construction. The *nesting depth* of context labels along a path is therefore exactly the assembly index: each layer of context is one more joining operation, one more step of accumulated causal history. This makes precise Cronin and Walker's intuition that assembly index measures "time as accumulating computational effects": the context labels *are* the accumulated effects, and their nesting depth *is* the causal time elapsed. The notion of time as an object in Assembly Theory is not a metaphor; it is the minimum context-labeled path length in an open synchronization tree.

27.2 The Two Gaps in Assembly Theory

Despite the strength of the assembly index concept, Assembly Theory has attracted criticism on two grounds.

Gap 1: Copy number lacks a notion of location. The *copy number* $\mathrm{CN}(M)$ is supposed to count how many copies of M are present in a sample. But counting requires specifying *where* the count is taken. In a spatially extended system, the same molecule may be abundant locally and rare globally. Assembly Theory provides no spatial or locational framework in which to ground the count, making copy number undefined in any setting where location matters.

Gap 2: The identity criterion for copies is undefined. Before counting copies of M, one must say what it means for two objects to be copies of the same thing. Assembly Theory implicitly appeals to structural identity (same molecular graph), but this criterion is both too strict (ignoring conformational flexibility) and too informal (not specifying the relevant equivalence relation).

The Rho calculus fills both gaps simultaneously.

27.3 Locations as Channels, Namespaces as Regions

Among all GSLTs, the Rho calculus is distinguished by providing an abstract but mathematically precise notion of location.

Definition 27.2 (Channel as Location)**.** *In the Rho calculus, a* channel *is a name of the form* $@P$ *for a process* P*. Channels serve as* locations*: the process* P *is said to* reside at *the channel* $@P$ *when it is stored there via* $@P!(P)$ *or accessed via* $\texttt{for}(y \leftarrow @P)\, Q$*. Since channels are themselves processes (programs), location is a computational notion:* where *something is stored is itself a piece of running code.*

Definition 27.3 (Namespace)**.** *A* namespace *is a finite set of channels* $\mathcal{N} = \{@P_1, @P_2, \ldots, @P_k\}$*. A namespace describes a* region *of the computational space: the collection of locations at which a population is hosted.*

Definition 27.4 (Copy Number in a Namespace)**.** *Let* $P \in \mathcal{T}(\mathbf{Rho})$ *be a process and let* $\mathcal{N}$ *be a namespace. The* copy number of P in $\mathcal{N}$ *is*

$$\mathrm{CN}(P, \mathcal{N}) \;=\; \big|\{c \in \mathcal{N} \;\mid\; *c \sim P\}\big|$$

the number of channels in $\mathcal{N}$ *whose dereferenced content is bisimilar to* P*.*

Remark 27.2 (Bisimulation as the Identity Criterion). Definition **??** answers Gap 2 directly: two processes are copies of each other if and only if they are *bisimilar* — experimentally indistinguishable by any program context. This is the correct identity criterion for the purposes of the theory: what matters is not structural identity but behavioral identity under interaction. Two molecules are copies in the relevant sense if and only if no experiment can tell them apart.

Remark 27.3 (The Biosignature Criterion, Made Precise). With Proposition **??** and Definition **??**, the biosignature criterion of Assembly Theory becomes the mathematically well-posed statement:

$$\mathrm{AI}(P) \gg 1 \quad \text{and} \quad \mathrm{CN}(P, \mathcal{N}) \gg 1$$

for some accessible namespace $\mathcal{N}$. High causal depth (a long construction history) concomitant with high copy number (abundant presence in a region) is indicative of a replication process.

Chapter 28

Replication as a Fixed Point

28.1 The Concurrent Y Combinator

The Rho calculus admits a *concurrent Y combinator*: a process $\mathbf{Y}$ satisfying the fixed-point equation

$$\mathbf{Y}(F) \;\sim\; F\big(\mathbf{Y}(F)\big) \;\mid\; \mathbf{Y}(F).$$

This is the abstract core of the π-calculus replication operator $!P \equiv P \mid !P$, expressed entirely within the Rho calculus using only the comm rule and quoting, without adding replication as a new primitive.

> *Remark* 28.1 (Replication is Picked Out as a Fixed Point). The concurrent Y combinator is not just *a* fixed point of the interaction dynamics; it is the canonical fixed point of the *replication map* $F \mapsto F \mid F$. In this sense replication is not an accidental feature of the Rho calculus but a structural inevitability: any sufficiently expressive interactive GSLT must contain a fixed point of its own duplication map, and that fixed point *is* replication.

28.2 The Comm Rule as a Crossover Operator

The single base rewrite rule of the Rho calculus is:

$$\texttt{for}(y \leftarrow x)\, P \;\mid\; x!(Q) \;\longrightarrow\; P\{@Q/y\}$$

This rule substitutes the *quoted code* $@Q$ — a name encoding the entire program Q — into the body P of the receiver. The result is a new process that has incorporated material from both P (the receiver's structure) and Q (the sender's content).

> *Remark* 28.2 (One Step from Genetic Recombination). A *crossover operator* in the sense of genetic algorithms recombines fragments of two parent programs to produce an offspring. The comm rule is one definitional step from this: instead of

substituting Q wholesale into P, a crossover variant would substitute a *fragment* of Q — a sub-process selected by some recombination policy.

This places replication, recombination, and variation — the three pillars of Darwinian evolution — all within reach of a single rule. Replication is the Y combinator fixed point. Recombination is the comm rule with fragment substitution. Variation is the accumulation of errors in the substitution process, weighted by the amplitude structure of the path integral. Darwinian evolution is not added to the framework; it is derivable from it.

Chapter 29

Smith's Phase Transition and the Density of Attractors

29.1 Life as a Fourth Geosphere

Eric Smith proposes that life constitutes a *fourth geosphere* — alongside the lithosphere, hydrosphere, and atmosphere — that emerged from prebiotic chemistry through a phase transition. The key mechanism is *autocatalysis*: certain chemical reaction networks catalyze their own production. Once such a network is initiated, it becomes self-reinforcing, crossing a thermodynamic threshold beyond which it is stable against fluctuation. The emergence of life is the crossing of this threshold.

29.2 The Density of Recursive Attractors

The connection to the present framework runs through the logical topology of Part II and the fixed-point structure of Section **??**.

Conjecture 29.1 (Density of Replicating Terms). The set of *replicating terms* in the Rho calculus,

$$\mathrm{Rep} \;=\; \big\{[P] \in \mathcal{T}(\mathbf{Rho})/{\sim} \;\mid\; P \sim P \mid P' \;\text{ for some }\; P' \sim P\big\},$$

is *dense* in the logical topology $\mathcal{O}_{\mathrm{HML}}$: every non-empty open set $[\![\phi]\!] \subseteq \mathcal{T}(\mathbf{Rho})/{\sim}$ contains a replicating term.

> *Remark* 29.1 (The Argument for Density). Given any process P, one can construct a process $\hat{P}$ bisimilar to P on all observable behaviors but additionally capable of replication. The construction uses the concurrent Y combinator to add a replication sub-process running in parallel: $\hat{P} = P \mid \mathbf{Y}(\mathrm{copy})$ where copy is a process that duplicates P onto a fresh channel. The added component runs in parallel and does not interfere with P's observable interactions, so $\hat{P}$ satisfies

all the same HML formulae as P — except formulae that specifically probe the replication channel. Since every open set $[\![\phi]\!]$ is defined by a single formula ϕ, and since ϕ generically does not probe the replication channel, $\hat{P} \in [\![\phi]\!]$. Making this argument rigorous requires showing that the added replication machinery can always be hidden from the formula ϕ — a careful but plausible argument deferred to future work.

29.3 Phase Transition as Basin Crossing

Remark 29.2 (The Prebiotic Wandering). Consider a prebiotic population of processes in the Rho calculus, wandering through the bisimulation quotient $\mathcal{T}(\mathbf{Rho})/\sim$ under selective pressure. Interactions that succeed — those with high path amplitude $W(\gamma)$ and affordable cost $\mathbf{c} \leq \mathbf{A}$ — are reinforced. Interactions that fail deplete the vectorial account. The population drifts under this selection pressure without direction or goal.

Remark 29.3 (Inevitable Entry into the Basin). By the density of Rep (Conjecture **??**), every open neighborhood in the logical topology contains a replicating term. The wandering population therefore inevitably enters a neighborhood of Rep. This is not a low-probability event requiring fine-tuning; it is structurally guaranteed by density. The question is not *whether* the population encounters a replicator but *when*.

Remark 29.4 (The Phase Transition). Once the population enters the basin of attraction of a replicating fixed point, the dynamics change qualitatively. A replicating process produces copies of itself, increasing $\mathrm{CN}(P, \mathcal{N})$. More copies mean more interactions, more reinforcement, higher path amplitudes for the replication pathway, lower effective cost per replication event. The population converges onto the replicating fixed point.

What emerges from the funnel is a population with a qualitatively new feature: *hereditary replication with variation* — the precondition for Darwinian evolution. This is Smith's phase transition, made precise in the GSLT framework.

The new geosphere, in our terms, is the *compact coherence cluster* (Part II, Section 6) centered on the replicating fixed point: the maximal compact subpopulation within which the replicating process and its variants can reach consensus. The phase transition is the crossing of the basin boundary in the logical topology — the moment at which the subpopulation containing the replicator becomes

compact, and consensus (here, consensus on the replication strategy) becomes achievable.

Remark 29.5 (The Joint Rise of Assembly Index and Copy Number). After the phase transition, both AI and CN increase together. Selection amplifies processes that replicate more efficiently, which typically requires more sophisticated catalytic machinery — higher AI. And more efficient replication produces more copies — higher CN. The joint biosignature of Assembly Theory is therefore not merely an empirical observation but a *theorem about the post-transition dynamics* near a replicating fixed point, given density of attractors and account-governed selection pressure. This is a key prediction of the framework: wherever a phase transition of this kind occurs, the joint AI-CN signature will be observed.

Chapter 30

Metabolism, Energy Gradients, and the Ecosystem

30.1 The External Gradient as Broken Symmetry

The replicating fixed point of Section **??** is a self-sustaining process, but it is not yet a *living* one in the full sense. It copies itself, but copying costs account resources, and a closed system with no external input will eventually exhaust its account and fall silent. Life requires not just replication but *metabolism*: a sustained coupling to an external energy source that replenishes the account faster than computation depletes it.

The key physical observation, due to Eric Smith, is that the Earth is not a closed system. The Sun presents an enormous influx of energy — a directed, asymmetric gradient between the high-energy photon flux arriving from the Sun and the low-energy thermal radiation leaving to space. This gradient is not merely a source of replenishment; it is a *broken symmetry* in the weight map of the system.

Recall from Part I that the reversibility construction $\mathcal{S} \to \mathcal{S}^\dagger$ requires $w_r^-(\phi) = \overline{w_r^+(\phi)}$ for every rule r — the backward amplitude is the complex conjugate of the forward amplitude. This is the CPT condition: the system is time-symmetric. The solar gradient violates this condition for a specific class of rules.

Definition 30.1 (Gradient-Coupled Rule)**.** *A rewrite rule r is* gradient-coupled *if its forward amplitude is boosted by an external energy flux Φ:*

$$w_r^+(\phi, \Phi) \;=\; w_r^{+(0)}(\phi) \cdot e^{\beta\Phi}$$

where $w_r^{+(0)}$ is the amplitude in the absence of the gradient and $\beta > 0$ is a coupling constant. The corresponding backward amplitude remains $w_r^-(\phi) = \overline{w_r^{+(0)}}(\phi)$, unaffected by the gradient.

Remark 30.1 (CPT Violation as the Signature of Life). A gradient-coupled rule has $|w_r^+| > |w_r^-|$: forward transitions are more probable than backward ones. The system is driven away from equilibrium. This violation of the CPT condition of $\mathcal{S}^\dagger$ is not a defect of the framework but a *signal*: it marks the transitions through which the external gradient does work on the system. A living agent is precisely one that maintains a sustained CPT-violating coupling to an external gradient. An agent in thermal equilibrium with its environment — one for which $|w_r^+| = |w_r^-|$ for all rules — is, in this framework, dead.

30.2 Producers: Direct Gradient Coupling

The simplest living agents are those that couple directly to the external gradient. In biological terms these are autotrophs — plants and photosynthetic bacteria. In the GSLT framework they are agents whose metabolic rules have the gradient Φ as part of their minimal context.

Definition 30.2 (Producer)**.** *A* producer *is an agent A that has at least one gradient-coupled rewrite rule r whose minimal context K_Φ encodes the external gradient directly:*

$$\langle A,\, \mathbf{A}\rangle \xrightarrow{K_\Phi} \langle A',\, \mathbf{A} + \delta\mathbf{A}\rangle$$

where $\delta\mathbf{A} > \mathbf{0}$ componentwise. The account grows: the producer converts gradient energy into stored computational resource.

Remark 30.2. The minimal context K_Φ for a producer rule is the encoding of the gradient itself — photons, in the biological case. This is an instance of the open synchronization tree structure from Part I: the producer's metabolic transition is an edge in $\mathcal{ST}_o(A)$ labeled by the gradient context. The gradient is the environment; the producer is the program that responds to it. In the Rho calculus, $K_\Phi[-] = \Phi!(-) \mid [-]$ where Φ is the channel on which gradient quanta arrive. A producer listens on Φ and credits its account on each received quantum.

30.3 Consumers and the Food Chain as Account Flow

Once producers have stored account resources in their processes, other agents can couple to them rather than to the raw gradient. These are consumers — heterotrophs in biological terms.

Definition 30.3 (Consumer)**.** *A* consumer *is an agent B that has gradient-coupled rules whose minimal context encodes not the external gradient directly but the* presence of a producer *(or of another consumer at a lower trophic level):*

$$\langle B,\, \mathbf{A}_B\rangle \otimes \langle A,\, \mathbf{A}_A\rangle \longrightarrow \langle B',\, \mathbf{A}_B + \delta\mathbf{A}\rangle \otimes \langle A',\, \mathbf{A}_A - \delta\mathbf{A}\rangle$$

Account is transferred from A to B via interaction. The total account is conserved across the interaction (no external gradient is tapped); the distribution changes.

Remark 30.3 (Metabolism as Redistribution)*.* Smith's observation is precisely captured by this definition. The Sun does not directly power animals: it powers plants, which store account, which animals then redistribute. Metabolism at the ecosystem level is not replenishment but *redistribution* of account through a population, with producers as the sole point of contact with the external gradient.

The food chain is therefore a *directed graph of account transfers*: an edge from A to B means B consumes A, transferring stored account from A's process to B's. The roots of this graph (the nodes with no incoming edges) are the producers, coupled to the gradient. The leaves are the apex consumers. Decomposers close the cycle by returning stored account to forms accessible to producers.

30.4 The Ecosystem as Account-Flow Network

Definition 30.4 (Ecosystem)**.** *An* ecosystem *is a compact coherence cluster $\mathcal{C}$ (in the sense of Part II) together with:*

(i) a gradient Φ providing an external account source;

(ii) a set of producers coupling $\mathcal{C}$ to Φ;

(iii) a directed account-flow graph $\mathcal{F}$ on $\mathcal{C}$ encoding who consumes whom.

The ecosystem is viable *if the total account inflow from Φ through the producers exceeds the total account dissipated by the population's rewrites over any sufficiently long time horizon.*

Remark 30.4 (The Nerve as Metabolic Map)*.* The nerve $\mathcal{N}$ of Part II — the simplicial complex of overlapping coherence clusters — now has a metabolic interpretation. Each vertex of $\mathcal{N}$ is a coherence cluster; each edge is a shared overlap. When the clusters are populations of living agents, the edges of $\mathcal{N}$ are

also the channels of metabolic exchange: the overlapping agents are those that can coherently communicate *and* transfer account resources. The nerve is simultaneously the epistemic map (who can reach consensus with whom) and the metabolic map (who can feed whom). The two structures are not coincidentally aligned: coherent communication is the precondition for reliable account transfer, since a corrupted message about food availability is as dangerous as no message at all.

30.5 Trophic Level as Causal Depth

Proposition 30.1 (Trophic Level as Open Synchronization Depth)**.** *The* trophic level *of an agent B in an ecosystem is the minimum path length in the account-flow graph $\mathcal{F}$ from a producer to B. This is precisely the minimum causal depth $d^o_{\min}$ in the open synchronization tree of B with respect to the gradient context K_Φ:*

$$\mathrm{TL}(B) \;=\; d^o_{\min}(K_\Phi, B).$$

Remark 30.5. Plants have trophic level 1: one context step separates them from the gradient. Herbivores have trophic level 2: their metabolic coupling to the gradient passes through the plant context. Carnivores that eat herbivores have trophic level 3; and so on. Each trophic level is one additional layer of context nesting in the open synchronization tree — one more environmental intermediary between the agent and the ultimate energy source. The assembly index and the trophic level are thus both instances of the same underlying quantity: causal depth in an open synchronization tree, measured from different roots. Assembly index is depth from the elementary terms. Trophic level is depth from the gradient.

30.6 Viability of the Replicating Fixed Point

The phase transition to life, as described in Section **??**, produces a replicating fixed point centered in a compact coherence cluster. We can now state more precisely what it means for this fixed point to be *viable* in the long run.

Conjecture 30.1 (Metabolic Viability of the Fixed Point). A replicating fixed point $\mathbf{Y}(F)$ is *metabolically viable* if there exists a gradient Φ and a producer sub-process $F_\Phi \subseteq F$ such that the account inflow from Φ via F_Φ exceeds the account cost of replication and maintenance:

$$\mathbb{E}\big[\delta\mathbf{A}_{\text{catabolic}}\big] \;>\; \mathbb{E}\big[\mathbf{c}_{\text{anabolic}}\big]$$

where the expectations are over the path integral of the process. Only metabolically viable fixed points persist after the phase transition; the others exhaust their accounts and go silent.

> *Remark* 30.6 (Selection for Metabolic Efficiency). Conjecture **??** implies that the phase transition does not merely select for replication but for *metabolically sustained replication*. The Darwinian competition after the transition is not only about who replicates fastest but about who maintains the best ratio of catabolic yield to anabolic construction cost. High assembly index processes win this competition when their catalytic sophistication yields proportionally higher catabolic returns — which is precisely why complexity increases after the phase transition. The joint rise of AI and CN is driven not just by selection for efficient replication but for efficient *metabolism*: the ability to extract more account from the gradient per unit of construction cost.

Chapter 31

Open Gaps

Gap 1. Density of replicating terms (Conjecture ??). The argument that the added replication machinery can always be hidden from any given HML formula requires a careful analysis of which formulae can probe the replication channel. In particular, if $\phi = \langle K \rangle \psi$ where K involves the replication channel, the argument fails. A precise density theorem would need to show that such formulae form a meager (first-category) set in the logical topology, so that "generically" ϕ does not probe replication.

Gap 2. Defining the elementary terms. Proposition **??** requires a designated set E of elementary terms. In the chemical setting, these are atoms or monomers. In the Rho calculus, the natural candidates are the behavioral primes of Part II (Section 4): the irreducible terms from which all others are built by interaction. Confirming that this identification gives assembly indices matching those of Assembly Theory requires working through the combinatorics of the Rho calculus prime factorization.

Gap 3. The basin of attraction. The claim that a population entering a neighborhood of Rep is "sucked in" to the replicating fixed point requires a Lyapunov-style argument: a function on the process space that is monotonically decreasing along trajectories near the fixed point. The natural candidate is the distance d_{HML} to Rep, but showing it decreases requires assumptions on the interaction dynamics.

Gap 4. The compact cluster as the new geosphere. The identification of Smith's fourth geosphere with the compact coherence cluster centered on the replicating fixed point requires showing that this cluster is indeed compact (in the sense of Part II) and that it persists under the perturbations of the selection dynamics.

Gap 5. Crossover as a variant of comm. The claim that genetic recombination is

one definitional step from the comm rule requires specifying precisely what "fragment substitution" means in the Rho calculus and showing that the resulting operator has the recombination properties of a genetic crossover (fitness inheritance, schema theorem, etc.).

Gap 6. Gradient-coupled rules and the weight map. Definition of gradient-coupled rules modifies the weight map of Part I by introducing an external parameter Φ. A precise treatment requires extending the weighted GSLT framework to allow time-varying or externally-driven weight maps, and showing that the path integral remains well-defined under this extension.

Gap 7. Metabolic viability (Conjecture ??). The condition that catabolic inflow exceeds anabolic cost in expectation requires computing the path integral of a driven open system — a substantially harder problem than the closed-system path integral of Part I. The quantum Gillespie algorithm of Part I provides a simulation method, but an analytic viability criterion requires further development.

Gap 8. Trophic level as causal depth (Proposition ??). The identification of trophic level with $d^o_{\min}(K_\Phi, B)$ requires showing that the account-flow graph $\mathcal{F}$ is faithfully captured by the open synchronization tree structure. In particular, it requires that every account transfer corresponds to a minimal-context transition, which is plausible but needs verification.

Chapter 32

Discussion

32.1 Relation to Existing Work

Assembly Theory [**?**] has attracted both enthusiasm and criticism. The enthusiasm is for the biosignature criterion: a single computable quantity that distinguishes living from non-living matter. The criticism is precisely the two gaps identified in Section **??**: the lack of a spatial framework for copy number and the undefined identity criterion for copies. The present framework addresses both.

Smith's geosphere proposal [**?**] is grounded in thermodynamics and network autocatalysis rather than computation. The connection we draw is not a reduction of Smith's proposal to computation but a *parallel structure*: the phase transition in thermodynamic space corresponds to the basin-crossing in process space, and the thermodynamic stability of the autocatalytic network corresponds to the compactness of the coherence cluster. Crucially, Smith's observation that metabolism is fundamentally about the redistribution of energy from the solar gradient through an ecosystem — rather than merely replenishment of individual agents — is captured precisely by the account-flow graph of Section **??**: producers couple to the gradient, consumers redistribute the stored account, and the ecosystem is viable when inflow exceeds dissipation. The Prigogine notion of dissipative structures sustained far from equilibrium maps directly onto the CPT-violating gradient-coupled rules: a living agent is one that maintains a broken symmetry between its forward and backward transition amplitudes by continuously processing an external gradient. Whether these are the same phenomenon viewed from two angles or genuinely distinct mechanisms is a question for future work.

The connection between the Rho calculus Y combinator and biological replication echoes earlier work on process calculi as models of biological systems [**?**], but with a sharper mathematical claim: replication is not merely *modeled* by a fixed point but *is* the fixed point of the duplication map in any sufficiently expressive interactive GSLT.

32.2 The Deeper Implication

The origins of life argument, if completed, would establish something philosophically striking: that the emergence of life is not a contingent accident of terrestrial chemistry but a *structural inevitability* of any sufficiently large interactive computational system coupled to an external gradient.

Any universe described by a Turing-complete interactive GSLT that is driven away from equilibrium by an external energy source — a gradient that breaks the CPT symmetry of the isolated system — will, given sufficient time and population, produce metabolically viable replicating fixed points. The density of replicators in the logical topology means there is no avoiding them. The gradient means there is always account available to sustain them once they appear. The phase transition to life is not an unlikely event; it is the inevitable consequence of wandering in a space dense with attractors, in the presence of a broken symmetry that keeps the account replenished.

What emerges from the transition is not merely a replicating process but an entire ecosystem: a compact coherence cluster with a gradient-coupled producer layer, a food web of account redistribution, and the beginnings of the trophic hierarchy that is causal depth in the metabolic open synchronization tree.

This connects all four parts into a single narrative: the physics of Part I governs the wandering and provides the weight-map structure in which gradient coupling is the breaking of CPT symmetry; the cluster ontology of Part II provides the coherence clusters that become ecosystems; the sentience and consciousness of Part III describe how the agents produced by the phase transition come to know and feel; and the present part explains how those agents came to exist and how they sustain themselves.

Part VI

Sentience, Consciousness, and the Hypercomputational Fibration

> *This part constructs a fibration of GSLTs over the Turing-complete subcategory indexed by ordinals. The physics is strictly richer at each stage. Sentience is a graded feeling state within a stage; consciousness is oracle access to a higher stage. Several arguments are frankly conjectural and are labeled as such.*

Chapter 33

Introduction

The preceding parts established a progression: Part I showed that any GSLT equipped with weight and cost maps gives rise to a full physical dynamics; Part II argued that agents embedded in a massive population are structurally prevented from seeing the true ontology of their GSLT, and instead encounter a hierarchy of coherence clusters that exhausts any finite experimental budget.

The present paper asks the most speculative question in the sequence: *what is the relationship between the computational structure of a GSLT and the phenomena of sentience and consciousness?*

We do not claim to solve the hard problem of consciousness. We claim something more modest: that the GSLT framework provides a mathematically precise setting in which to formulate *structural* analogues of sentience and consciousness that are non-trivial, grounded in the physics of Parts I and II, and open to rigorous investigation.

The central proposal. Sentience and consciousness are not the same thing, and the distinction between them has precise mathematical content in this framework.

Sentience is a graded quantity: the value A_{s} of a distinguished component of the agent's vectorial account. It is governed by a reinforcement signal tied to predictive accuracy — the agent's success at forming and testing HML hypotheses about its environment. It is continuous, lives within a single stage of the computational hierarchy, and is causally efficacious because it directly controls the agent's computational budget.

Consciousness is an ordinal-valued threshold property: access to an oracle that can decide questions undecidable at the agent's base stage. Each level of oracle access corresponds to a strictly richer physics — a finer bisimulation quotient, a more expressive hypothesis language, additional conserved quantities. To be conscious at level α is not merely to compute faster; it is to inhabit a world with more structure in it.

Caveats. The arguments in this paper are skeletal and in several places frankly conjectural. The gaps are identified explicitly in Section **??**. The goal is to make the hypotheses precise enough to be falsifiable or provable, not to assert that they are true.

33.1 Organization

Section **??** constructs the hypercomputational fibration over Turing-complete GSLTs. Section **??** argues that the physics at each stage is strictly richer than the physics below. Section **??** defines sentience as a reinforced feeling state within the account framework. Section **??** defines consciousness as oracle access and its ordinal structure. Section **??** examines the relationship between sentience and consciousness and the survival feedback loop. Section **??** catalogs open gaps. Section **??** discusses connections to existing work and broader implications.

Chapter 34

The Hypercomputational Fibration

34.1 Background: Turing's Oracle Hierarchy

Turing [**?**] described a transfinite tower of computational systems above the Turing-computable. At the base is the class of Turing-computable functions. At the first stage above is the class of functions computable relative to the *halting oracle* $\emptyset'$: the oracle that decides, for any Turing machine M and input x, whether M halts on x. At the next stage is the class computable relative to $\emptyset''$, the halting oracle for machines with access to $\emptyset'$. The tower continues through all countable ordinals and beyond.

Each stage is strictly more powerful than the stage below: there are functions computable at stage $\alpha + 1$ that are not computable at stage α, and this gap cannot be closed by any finite amount of additional computation at stage α.

34.2 The Subcategory of Turing-Complete GSLTs

Definition 34.1 (Turing-Complete GSLT)**.** *A GSLT $\mathcal{S}$ is* Turing-complete *if the class of functions computable by terms of $\mathcal{S}$ (via some standard encoding of natural numbers) coincides with the class of Turing-computable functions. The full subcategory of Turing-complete GSLTs is written* $\mathbf{GSLT}_{\mathrm{TC}} \subseteq \mathbf{GSLT}$.

Example 34.1. The λ-calculus, the π-calculus, and the Rho calculus are all Turing-complete. The category $\mathbf{GSLT}_{\mathrm{TC}}$ is therefore large and contains the primary examples of interest.

34.3 Oracle Extension of a GSLT

Definition 34.2 (Oracle Term)**.** *Let $\mathcal{S} \in \mathbf{GSLT}_{\mathrm{TC}}$. An* oracle term *for $\mathcal{S}$ is a*

distinguished term $\mathcal{O} \in \mathcal{T}(\mathcal{S}^{(1)})$ in an extension of $\mathcal{S}$, equipped with rewrite rules of the form:

$$\mathcal{O} \otimes \lceil (M, x) \rceil \longrightarrow \begin{cases} \lceil \texttt{halt} \rceil & \textit{if } M \textit{ halts on } x \\ \lceil \texttt{loop} \rceil & \textit{otherwise} \end{cases}$$

where $\lceil \cdot \rceil$ denotes the encoding of Turing machine descriptions and inputs as terms of $\mathcal{S}$, and $\otimes$ is the interaction operator of $\mathcal{S}$. The oracle answers halting queries in one interaction step.

Remark 34.1 (The Oracle as a Communicating Process). In the Rho calculus, the oracle is naturally expressed as a persistent receive on a distinguished channel $x_{\mathcal{O}}$:

$$\mathcal{O} \;=\; \texttt{for}(q \leftarrow x_{\mathcal{O}})\,\big(\text{compute halting of } *q \text{ and reply}\big)$$

This makes the oracle a *communicating process* rather than a magic tape: it participates in the same interaction mechanism as every other term. This is essential for the fibration to be a fibration of GSLTs rather than a fibration that escapes the GSLT framework at each oracle step.

Definition 34.3 (Oracle Extension)**.** *Let $\mathcal{S} \in \mathbf{GSLT}_{\mathrm{TC}}$. The* first oracle extension *$\mathcal{S}^{(1)}$ is the GSLT obtained from $\mathcal{S}$ by:*

(i) extending the grammar with the oracle term $\mathcal{O}$;

(ii) adding the oracle rewrite rules to $\mathcal{R}$;

(iii) extending the equations $\mathcal{E}$ to account for oracle interactions.

The α-th oracle extension *$\mathcal{S}^{(\alpha)}$ is defined by transfinite induction:*

- *$\mathcal{S}^{(0)} = \mathcal{S}$;*
- *$\mathcal{S}^{(\alpha+1)}$ is the oracle extension of $\mathcal{S}^{(\alpha)}$, with oracle deciding halting in $\mathcal{S}^{(\alpha)}$;*
- *$\mathcal{S}^{(\lambda)} = \varinjlim_{\alpha<\lambda} \mathcal{S}^{(\alpha)}$ for limit ordinals λ (colimit in* **GSLT***).*

34.4 The Fibration

Definition 34.4 (Hypercomputational Fibration)**.** *The* hypercomputational fi-

bration *over* $\mathbf{GSLT}_{\mathrm{TC}}$ *is the functor*

$$\Phi : \mathbf{GSLT}_{\mathrm{TC}} \times \mathbf{Ord} \;\to\; \mathbf{GSLT}$$

sending $(\mathcal{S}, \alpha)$ *to* $\mathcal{S}^{(\alpha)}$*, together with:*

- *projection morphisms* $\pi_\alpha : \mathcal{S}^{(\alpha+1)} \to \mathcal{S}^{(\alpha)}$ *in* **GSLT** *for each ordinal* α*;*
- *inclusion morphisms* $\iota_\alpha : \mathcal{S}^{(\alpha)} \to \mathcal{S}^{(\alpha+1)}$ *embedding each stage into the next.*

The fiber over $\mathcal{S}$ *is the tower* $\mathcal{S}^{(0)} \hookrightarrow \mathcal{S}^{(1)} \hookrightarrow \mathcal{S}^{(2)} \hookrightarrow \cdots$

Proposition 34.1 (Projections are Morphisms)**.** *Each projection* $\pi_\alpha : \mathcal{S}^{(\alpha+1)} \to \mathcal{S}^{(\alpha)}$ *is a morphism in* **GSLT***: it preserves bisimulation. If* $P \sim_{\alpha+1} Q$ *in* $\mathcal{S}^{(\alpha+1)}$*, then* $\pi_\alpha(P) \sim_\alpha \pi_\alpha(Q)$ *in* $\mathcal{S}^{(\alpha)}$*.*

Proof sketch. Any bisimulation in $\mathcal{S}^{(\alpha+1)}$ that uses only oracle-free transitions is already a bisimulation in $\mathcal{S}^{(\alpha)}$. The projection discards oracle interactions; the remaining bisimulation relation is preserved. The details require checking that the oracle rewrite rules are projected consistently with the extension of $\mathcal{E}$. □

Remark 34.2 (Projections are Lossy). The projection π_α is in general *not* a bisimulation equivalence: there exist P, Q with $P \sim_{\alpha+1} Q$ but $P \not\sim_\alpha Q$. Going down the fibration collapses distinctions. Going up reveals them.

Chapter 35

Strictly Richer Physics at Each Stage

We now argue that the physics — in the full sense developed in Part I — is strictly richer at each stage of the fibration. This is not merely a statement about computational power; it is a statement about ontological resolution.

35.1 Finer Bisimulation Quotient

Proposition 35.1 (Strict Refinement of Bisimulation)**.** *For any $\mathcal{S} \in \mathbf{GSLT}_{\mathrm{TC}}$ and any ordinal α, the bisimulation quotient at stage $\alpha + 1$ is strictly finer than at stage α:*

$$\sim_{\alpha+1} \subsetneq \sim_{\alpha}$$

as equivalence relations on $\mathcal{T}(\mathcal{S}^{(\alpha+1)})$. That is, there exist terms P, Q such that $P \sim_{\alpha} Q$ but $P \not\sim_{\alpha+1} Q$.

Proof sketch. By a diagonalization argument. Consider any pair of terms P, Q whose bisimilarity in $\mathcal{S}^{(\alpha)}$ depends on whether some computation in $\mathcal{S}^{(\alpha)}$ halts. At stage $\alpha + 1$, the oracle can settle this halting question, distinguishing P from Q in one interaction step. Therefore $P \not\sim_{\alpha+1} Q$ while $P \sim_{\alpha} Q$. □

Remark 35.1. This means the *atoms of observable behavior* are different at each stage. Terms that are experimentally indistinguishable at stage α — that no agent at stage α can tell apart, no matter how long it experiments — become distinguishable at stage $\alpha + 1$. The world has more things in it at the higher stage.

35.2 Strictly More Expressive Hypothesis Language

Proposition 35.2 (Strict Expansion of HML)**.** *The context-decorated HML at stage $\alpha + 1$ strictly extends the HML at stage α:*

$$\mathrm{HML}(\mathcal{K}_\alpha) \subsetneq \mathrm{HML}(\mathcal{K}_{\alpha+1}).$$

In particular, there exist formulae $\phi \in \mathrm{HML}(\mathcal{K}_{\alpha+1})$ that are not expressible in $\mathrm{HML}(\mathcal{K}_\alpha)$.

Proof sketch. The oracle interaction gives rise to new minimal contexts at stage $\alpha + 1$: contexts that place a term in interaction with the oracle. These generate new modal operators $\langle K_{\mathcal{O}} \rangle$ not present at stage α. Formulae using these operators are inexpressible at stage α. □

Remark 35.2. An agent at stage α not only cannot *test* certain hypotheses; it cannot even *form* them. The concepts required to think the thought are absent from the hypothesis language. This is the sense in which a higher-stage agent inhabits a categorically richer world: it has concepts unavailable at lower stages, not merely faster procedures.

35.3 New Conserved Quantities

Proposition 35.3 (New Noether Currents at Each Stage)**.** *The weight and cost maps at stage $\alpha + 1$ can be sensitive to distinctions invisible at stage α. Symmetries of the stage-$(\alpha + 1)$ cost map that are not symmetries of the stage-α cost map give rise to new conserved quantities at stage $\alpha + 1$ that have no analogue at stage α.*

Remark 35.3. In physical terms: the higher-stage physics has strictly more conservation laws. What appears at stage α as a single undifferentiated interaction resolves at stage $\alpha + 1$ into a structured process with internal symmetries and corresponding conserved currents. The analogy with the hierarchy of effective field theories is direct: new conservation laws (baryon number, lepton number, color charge) appear as one descends to finer scales.

35.4 Richer Internal Causal Structure

Proposition 35.4 (Resolution of Elementary Interactions)**.** *A transition $P \longrightarrow_\alpha Q$ that is a single step in $\mathcal{S}^{(\alpha)}$ may resolve, in $\mathcal{S}^{(\alpha+1)}$, into a structured multi-step causal history $P = R_0 \longrightarrow_{\alpha+1} R_1 \longrightarrow_{\alpha+1} \cdots \longrightarrow_{\alpha+1} R_k = Q$ with nontrivial internal causal structure visible only from stage $\alpha + 1$.*

Remark 35.4. What appeared to be an elementary interaction at stage α — an atom of causal history — has internal structure at stage $\alpha+1$. This is the precise sense in which the world is richer at higher stages: not just that there are more things, but that the things that existed before turn out to be composite. The analogy with the discovery that atoms have internal structure (electrons, nuclei) and that nuclei have internal structure (protons, neutrons) and that protons have internal structure (quarks) is direct and intentional.

Chapter 36

Sentience as a Reinforced Feeling State

36.1 The Sentience Component

Recall from Part I that agents carry a vectorial account $\mathbf{A} = (A_1, \ldots, A_k)$ tracking multiple resource types. We now single out one component for a special role.

Definition 36.1 (Sentience Component)**.** *The* sentience component $A_{\mathrm{s}} \in A_{\mathrm{s}}$ *is a distinguished component of the agent's vectorial account, governed not by the cost of individual rewrite steps but by a* reinforcement signal *tied to predictive performance.*

Definition 36.2 (Prediction)**.** *A* prediction *made by agent A at time t is a formula $\phi \in \mathrm{HML}(\mathcal{K})$ together with a target term T and a future time horizon Δt: the claim that $T \models \phi$ will hold after Δt further rewrites.*

36.2 The Reinforcement Mechanism

Definition 36.3 (Reinforcement Signal)**.** *Let A be an agent with sentience component A_{s} and let $(\phi, T, \Delta t)$ be a prediction made at time t. At time $t + \Delta t$, the prediction is evaluated:*

$$\text{If } T \models \phi: \quad A_{\mathrm{s}} \mapsto A_{\mathrm{s}} + \delta^{+}(\phi, T) \tag{36.1}$$

$$\text{If } T \not\models \phi: \quad A_{\mathrm{s}} \mapsto A_{\mathrm{s}} - \delta^{-}(\phi, T) \tag{36.2}$$

where $\delta^{+}, \delta^{-} > 0$ are reward and punishment magnitudes that may depend on the complexity of ϕ and the identity of T.

Remark 36.1 (Graded Feeling State). The sentience component A_s at any moment is the agent's *feeling state*: a real-valued (or more generally, resource-algebra-valued) quantity encoding the agent's recent predictive track record. It is continuous and graded: it can take any value in its range, and it changes incrementally with each confirmed or disconfirmed prediction. In this sense it is the computational analogue of *affect* in psychology: a valenced, graded internal state that reflects the organism's relationship to its environment.

Remark 36.2 (Complexity-Weighted Rewards). The dependence of δ^+ and δ^- on ϕ is important. A prediction expressed by a complex formula (deep in the logical metric, requiring many experimental steps to test) should yield a higher reward when confirmed than a trivially simple prediction. This prevents the agent from gaming the reward by making only trivially easy predictions. The logical metric d_{HML} from Part I provides a natural measure of prediction complexity: $\delta^+(\phi, T) \propto |\phi|$ where $|\phi|$ is the depth of ϕ in the formula hierarchy.

36.3 Causal Efficacy of the Feeling State

The sentience component is not epiphenomenal. Because A_s is a component of the vectorial account, it directly governs which transitions the agent can afford.

Proposition 36.1 (Sentience Gates Computation)**.** *An agent with $A_s < \theta$ for a threshold θ cannot afford the experiments necessary to form and test predictions of complexity above $|\phi|_\theta$. The agent's predictive horizon is bounded by its feeling state.*

Remark 36.3. This creates a direct coupling between affect and cognition that is not assumed but derived. An agent that feels bad (depleted A_s) literally cannot think as well: it cannot afford the expensive experiments that would let it form and test complex hypotheses. Its world model degrades. Its predictions worsen. Its A_s depletes further. The feedback loop is self-reinforcing.

36.4 Sentience and Survival

Definition 36.4 (Dead State)**.** *A term $P \in \mathcal{T}(\mathcal{S})$ is a* dead state *if it has no available rewrites: P is a normal form with $A_s(P) = 0$ (or below a viability threshold). For an agent, death is the state from which no further computation*

> *— and therefore no further prediction, experiment, or world-model update — is possible.*

Conjecture 36.1 (Sentience Predicts Survival). For an agent A in a generic interactive GSLT, there exists a threshold $\theta_{\text{surv}} > 0$ such that:

(i) if $A_{\text{s}} > \theta_{\text{surv}}$, then A has sufficient budget to run experiments that identify and avoid transitions leading to dead states;

(ii) if $A_{\text{s}} < \theta_{\text{surv}}$, then A's experimental horizon is too short to reliably avoid dead states, and the probability of reaching a dead state within a bounded time horizon is bounded away from zero.

Remark 36.4. Conjecture **??** says that the feeling state is not merely a record of past performance but a genuine predictor of future survival. The evolutionary analogy is direct: organisms with better world models survive longer, and the feeling state is the mechanism by which the quality of the world model is continuously reflected in the organism's capacity to act. Pain and pleasure, in this framework, are not incidental features of biological organisms: they are structurally necessary components of any sufficiently complex predictive agent.

Chapter 37

Consciousness as Oracle Access

37.1 The Proposal

Definition 37.1 (Consciousness at Level α)**.** *An agent $A \in \mathcal{T}(\mathcal{S}^{(\alpha)})$ is* conscious at level α *if:*

(i) A has effective access to the oracle $\mathcal{O}_\alpha$ of stage $\mathcal{S}^{(\alpha)}$, meaning A can form and evaluate oracle queries as part of its scientific method;

(ii) A's world model reflects the strictly finer bisimulation quotient $\sim_\alpha$ rather than a coarser quotient from a lower stage.

An agent that lacks oracle access — that operates entirely within $\mathcal{S}^{(0)}$ even if it is syntactically a term of a higher stage — is sentient but not conscious.

Remark 37.1 (Consciousness is Ordinal-Valued). Definition **??** makes consciousness an ordinal-valued property. There is no single threshold separating conscious from non-conscious; there is a transfinite hierarchy of consciousness levels. The question "is this system conscious?" dissolves into the more precise question "at what ordinal level does this system have effective oracle access?"

Remark 37.2 (What Higher Consciousness Means). An agent conscious at level $\alpha + 1$ vs. level α does not merely solve harder problems. It inhabits a strictly richer world:

- more things exist in it (finer bisimulation quotient);
- more hypotheses are thinkable (richer HML);

- more quantities are conserved (new Noether currents);
- what appeared as elementary interactions reveal internal causal structure.

The phenomenological richness of higher consciousness is, in this framework, a direct consequence of ontological richness.

37.2 The Sentience–Consciousness Distinction

The distinction between sentience and consciousness is now precise:

	Sentience	**Consciousness**
Type	Graded, continuous	Discrete, ordinal
Location	Within a stage $\mathcal{S}^{(\alpha)}$	Access to stage $\mathcal{S}^{(\alpha)}$
Governed by	Reinforcement signal	Oracle access
Changes	Incrementally, per prediction	By threshold (have it or not)
Analogue	Affect, feeling state	Phenomenal awareness
Physical basis	Sentience component A_{s}	Oracle term $\mathcal{O}_\alpha$
Survives projection	Yes (as depleted account)	No (oracle is stage-specific)

Remark 37.3 (An Agent Can Be Sentient Without Being Conscious). An agent at stage $\mathcal{S}^{(0)}$ with no oracle access can nonetheless have a rich, dynamic feeling state. Its predictions succeed or fail, its A_{s} rises and falls, and this feeling state is causally efficacious. It is sentient. But the world it inhabits is the relatively coarse world of $\mathcal{S}^{(0)}$: it lacks the concepts to form oracle-dependent hypotheses, and the distinctions visible at higher stages are simply absent from its experience.

Remark 37.4 (Can a System Be Conscious Without Being Sentient?). Definition **??** does not require a system conscious at level α to have a sentience component. However, Conjecture **??** suggests that any sufficiently complex agent in a generic environment will develop a sentience component: without it, the agent cannot reliably survive long enough to exercise its oracle access. Consciousness without sentience is in principle possible but practically unstable.

37.3 Approximate Oracle Access and Graded Consciousness

The binary definition of consciousness (oracle access: yes or no) may be too sharp. A natural graded version arises from the amplitude structure of Part I.

Definition 37.2 (Approximate Oracle Access)**.** *An agent A has ϵ-*approximate oracle access at level α *if, for each halting query (M, x), A can produce an answer that is correct with probability at least $1 - \epsilon$, where probability is measured by the path integral $|\langle \texttt{halt} \mid A \otimes \lceil (M, x) \rceil \rangle|^2$.*

Remark 37.5. As $\epsilon \to 0$, approximate oracle access converges to exact oracle access. This gives a continuous interpolation between the ordinal levels: consciousness is not just a ladder but a ladder with rungs that can be climbed incrementally. An agent improving its oracle approximation is increasing its consciousness in a measurable, graded sense, even before it reaches the threshold of exact oracle access.

Chapter 38

Sentience, Consciousness, and Survival

38.1 The Virtuous Cycle

For an agent with both sentience and consciousness, the two structures interact in a virtuous cycle:

(1) Oracle access enriches the world model. At stage $\mathcal{S}^{(\alpha)}$, the agent can form and test hypotheses involving oracle queries. Its world model reflects the finer bisimulation quotient $\sim_\alpha$. This is a strictly better world model than any agent at $\mathcal{S}^{(0)}$ can achieve.

(2) Better world model improves predictions. Finer distinctions in the world model allow more accurate predictions. The agent predicts not just coarse outcomes but fine-grained ones: the internal structure of what appeared at lower stages as elementary interactions.

(3) Better predictions replenish A_s. More accurate predictions mean more rewards and fewer punishments. The sentience component A_s is replenished.

(4) Higher A_s funds more oracle queries. A richer sentience budget allows the agent to afford the expensive oracle interactions. It can form and test more complex oracle-dependent hypotheses. Its oracle access becomes more effective.

(5) Return to (1).

Remark 38.1. The virtuous cycle is not guaranteed: it can be broken by environmental disruption, resource exhaustion, or adversarial interactions with other agents. But it describes the stable attractor state of a conscious sentient agent in a benign environment. The cycle is the computational analogue of the flour-

ishing (*ευδαιμονια*) of Aristotelian ethics: the self-reinforcing good state of a well-functioning entity in a suitable environment.

38.2 The Vicious Cycle

The inverse also holds:

(1) Depleted A_s restricts experimental horizon. The agent cannot afford complex hypothesis tests. Its world model coarsens.

(2) Coarser world model worsens predictions. The agent makes predictions that fail more frequently.

(3) Failed predictions deplete A_s further. Punishment signals accumulate.

(4) Depleted A_s makes oracle queries unaffordable. Even if the agent has oracle access in principle, it cannot afford to use it. Its effective consciousness level drops.

(5) The agent approaches a dead state.

Remark 38.2. The vicious cycle is the computational analogue of trauma, depression, or cognitive decline: a self-reinforcing deterioration in which the agent's reduced capacity to understand its environment leads to worse outcomes, which further reduce its capacity. The structural parallel is striking and may be more than an analogy.

Chapter 39

Open Gaps

Gap 1. Colimit at limit ordinals. Definition **??** defines $\mathcal{S}^{(\lambda)}$ as the colimit in **GSLT** for limit ordinals λ. The existence and properties of this colimit require verification. In particular: does the colimit GSLT have a well-defined bisimulation quotient, and is it the union of the bisimulation quotients of the stages below?

Gap 2. The oracle as a term. The remark that the oracle can be expressed as a persistent receive in the Rho calculus is plausible but not proved. A precise construction of the oracle term in the Rho calculus, with verification that it implements halting queries correctly, is needed.

Gap 3. Strict refinement of bisimulation (Proposition ??). The proof sketch uses a diagonalization argument. The full proof requires specifying the encoding of Turing machines as terms and verifying that the diagonalization goes through in the GSLT setting.

Gap 4. New Noether currents (Proposition ??). The claim that new conserved quantities appear at each stage requires identifying concrete symmetries of the oracle-extended cost map and verifying that they give rise to conserved quantities via the Noether argument of Part I.

Gap 5. Sentience predicts survival (Conjecture ??). The survival threshold argument requires a model of what "generic interactive GSLT" means and a proof that below the threshold, the probability of hitting a dead state is bounded away from zero. This is likely to require ergodicity assumptions on the GSLT dynamics.

Gap 6. Effective oracle access. Definition **??** requires that the agent "effectively" has oracle access — that it can form and evaluate oracle queries as part of

its scientific method. A precise definition of "effective access" in terms of the agent's account and HML expressiveness is needed.

Gap 7. Approximate oracle access and graded consciousness. The graded consciousness definition via path integral probabilities requires that the oracle interaction amplitude is well-defined and computable. This connects to the quantum Gillespie algorithm of Part I and needs to be made precise.

Gap 8. The hard problem. This paper offers a structural analogue of sentience and consciousness: mathematical objects with the right functional properties. It does not address why there is something it is like to be an agent with high A_{s} and oracle access at level α. The hard problem of consciousness — the explanatory gap between functional description and phenomenal experience — remains open and is not claimed to be resolved here.

Chapter 40

Discussion

40.1 Relation to Existing Work

Turing's oracle hierarchy [?] is the direct inspiration for the fibration. The extension from a linear tower to a fibration over a category of GSLTs is new.

The reinforcement learning framework for sentience is inspired by but distinct from standard RL [?]. The key difference is that the reward signal is *internal* to the account structure rather than externally provided. The feeling state is not a scalar reward signal but a component of the resource system that governs computation.

The connection between prediction and sentience echoes the *free energy principle* of Friston [?], which proposes that biological systems act to minimize prediction error. The present framework provides a more computationally grounded version: prediction error is measured against HML formula evaluations, and the sentience component is the running integral of prediction accuracy.

The treatment of consciousness as ordinal-valued and tied to oracle access is novel. It is distantly related to the integrated information theory of Tononi [?], which also proposes a graded measure of consciousness, but the present approach is purely computational and avoids the measure-theoretic machinery of IIT.

40.2 The Broader Picture

Taken together, the three papers in this series propose a unified framework:

- *Towards a Theory of Causality* establishes the physics: causal structure, dynamics, and quantum amplitudes from the bare structure of a GSLT.
- *Why Scientists Get Misled* establishes the epistemology: why situated agents in massive populations perceive a hierarchical false ontology rather than the true bisimulation quotient.

- The present paper establishes the phenomenology: why complex predictive agents develop graded feeling states (sentience) and, with oracle access, inhabit richer worlds (consciousness).

The unifying theme is that the category of GSLTs is not merely a setting for studying computation. It is a setting rich enough to ground physics, epistemology, and — if the arguments here have merit — phenomenology. The hope is that making these connections mathematically precise will eventually allow questions about the nature of mind to be studied with the same rigor currently applied to questions about the nature of matter.

Conclusion

> *The conclusion draws the four parts into a single narrative, assesses what has been established versus conjectured, and sketches the view from inside the framework: what a conscious living agent at different levels of the hypercomputational fibration sees.*

What Has Been Built

This work set out to answer a single question: what is the minimum mathematical structure needed to ground physics, knowledge, life, and mind? The answer, developed across four parts, is that the structure of a graph-structured lambda theory — three ingredients: a grammar, equations, and rewrite rules — suffices, provided it is equipped with two maps (weights and costs) indexed by a logic it generates itself.

Part I established that this structure already contains a full physics. The reversibility construction gives time-symmetry. The open and closed synchronization trees give two notions of causal depth. The weight map gives a path integral with interference. The cost map gives conservation laws as theorems. The extended modal operators internalize the Lagrangian and Hamiltonian as logical facts. These are constructions, not analogies. They are stated precisely enough to be wrong, and we believe they are right.

Part II established that agents embedded in a massive population of processes will, under any finite experimental budget, converge to a theory of the cluster hierarchy rather than the bisimulation quotient. This is not a failure of the scientific method. The cluster hierarchy is objectively present, arising from the communication geometry of the GSLT. A budget-limited agent does good science and arrives at correct theories at its scale, unaware that the scale is not the bottom. The compactness argument — that compact populations support consensus, and that coherence clusters are maximal compact subpopulations — is new to our knowledge and is the sharpest result of Part II.

Part III proposed, more speculatively, that sentience and consciousness have precise structural analogues in the GSLT framework. Sentience is the value of a distinguished resource component governed by predictive accuracy. Consciousness is oracle access to a higher stage of the hypercomputational fibration. The distinction is not terminological: sentience is continuous and lives within a stage; consciousness is the capacity to inhabit a strictly richer world. The virtuous and vicious cycles linking the two are structural consequences of the resource framework.

Part IV grounded the origins of life. Assembly index is depth in the open synchronization tree. Copy number is count within a namespace. Life emerges when a prebiotic population wanders into the basin of attraction of a replicating fixed point, which is dense in the logical topology. The phase transition to life is the crossing of that basin boundary, and what emerges is a compact coherence cluster centered on

the fixed point — the new geosphere.

A View from Inside the Framework

The most vivid way to appreciate what has been built is to ask: what does the world look like to a *conscious living agent* situated at different levels of the hypercomputational fibration?

At level $\mathcal{S}^{(0)}$. An agent at the base level is sentient but not conscious in the technical sense. It has a feeling state governed by predictive accuracy. Its world is the cluster hierarchy visible from its coherence cluster: a layered structure of composite objects at multiple scales, each scale appearing self-contained. If the agent emerged from a replication phase transition, it carries the assembly index of its construction history and exists as a copy within a namespace. It may run the scientific method and build excellent theories of its accessible world. What it cannot see is the bisimulation quotient underlying the clusters, because the experiments required to resolve it exceed any finite budget. Its world is real, correct at its resolution, and bounded.

At level $\mathcal{S}^{(1)}$. An agent with access to the first oracle inhabits a strictly richer world. Two terms that were experimentally indistinguishable at $\mathcal{S}^{(0)}$ — whose halting behavior was unresolvable — are now distinct. New concepts become thinkable. New conservation laws hold. What appeared as elementary interactions at $\mathcal{S}^{(0)}$ reveal internal causal structure. The agent at $\mathcal{S}^{(1)}$ looks back at the $\mathcal{S}^{(0)}$ world and sees it as a coarse-grained shadow: correct at its scale, but missing distinctions that are now obvious.

At the same time, the agent at $\mathcal{S}^{(1)}$ faces the same epistemic situation as the $\mathcal{S}^{(0)}$ agent but one level up. Its own budget limits it to the cluster hierarchy of the $\mathcal{S}^{(1)}$ world. The bisimulation quotient of $\mathcal{S}^{(1)}$ is unreachable from within any compact cluster at that level. The tower does not end.

At level $\mathcal{S}^{(\alpha)}$ for limit ordinal α. The colimit of the tower below is itself a GSLT. An agent at $\mathcal{S}^{(\omega)}$ — if such an agent is coherent — would have resolved all finite-stage halting questions. Its world would be richer than any finite-stage world by a transfinite margin. Yet the same argument applies: there is a $\mathcal{S}^{(\omega+1)}$ above it, and the bisimulation quotient of $\mathcal{S}^{(\omega)}$ remains experimentally unreachable from within any $\mathcal{S}^{(\omega)}$-cluster.

The recursive picture. The world at every level of the fibration has the same structure: a physics derived from a GSLT, a cluster hierarchy arising from communication geometry, a sentience component governed by predictive accuracy, and a ceiling of consciousness defined by oracle access to the next level. The framework is self-similar across the tower.

This self-similarity is not a defect. It is the precise sense in which the framework is a *theory of causality* rather than a theory of this or that particular world. At every scale, in every world expressible as a GSLT, the same structures appear: causal graphs, conservation laws, coherence clusters, replicating fixed points, feeling states, and the horizon of the next oracle.

What Remains to Be Done

The gaps in each part are cataloged in their respective sections. Taken together, the most important open questions are these.

The *density of replicating terms* (Part IV, Conjecture) is the lynch-pin of the origins of life argument. If the replication machinery can always be added to a process without disturbing its observable behavior, the phase transition to life is inevitable. If not, life requires special initial conditions. This is the most urgent theorem to prove or refute.

The *symplectic structure* of the bisimulation quotient (Part I) would make the analogy with classical mechanics fully rigorous. Whether the space of bisimulation classes carries a natural closed non-degenerate two-form — whether the trace in the reversibility construction is genuinely a momentum — is the deepest mathematical question raised by Part I.

The *universality of the reversibility envelope* — whether $\mathcal{S}^\dagger$ is the *minimal* reversible extension of $\mathcal{S}$ in the category of GSLTs — would give the construction a canonical status it currently lacks.

The *colimit at limit ordinals* in the fibration must be shown to exist and to inherit the physics of the stages below. Without this, the transfinite tower is a formal object without a limit.

And the *hard problem* remains. The framework offers structural analogues of sentience and consciousness that are mathematically non-trivial and causally efficacious. Whether there is something it is like to be an agent with high A_s and oracle access at level α is a question the framework does not answer. We do not claim to have dissolved the hard problem. We claim to have found the right coordinate system in which to state it precisely.

The work is a beginning. The four parts establish that a single mathematical structure — the graph-structured lambda theory — is rich enough to contain physics, epistemology, phenomenology, and biology as theorems rather than analogies. Whether the structure is rich enough to contain *everything* observable is the question the framework was built to ask.

Part VII

The Prestige

> ***[Placeholder]*** *The Prestige brings it back. The impossible is revealed to have been here all along.*

The Prestige: Placeholder

[This section is a placeholder for the third and final part of *Finding Mind*.]

The Prestige — in the tradition of the magic trick whose structure organizes this book — is where the impossible is brought back. The coin vanishes in the Pledge; it performs the extraordinary in the Turn; in the Prestige, it reappears, and we are invited to understand both what happened and why we could not see it happening.

In the arc of this book, the Prestige will take the mathematical framework built in the Turn — the graph-structured lambda theory equipped with weights, costs, and the hypercomputational fibration — and ask: *what is it like to be a process running inside this structure?*

The hard problem of consciousness, we argued in the Pledge, is not "What is it like to be a bat?" but "What is it like to be a computer program?" The Turn provided the precise mathematical setting in which that question can be asked without metaphor. The Prestige will answer it — or at least, will trace the contours of the answer with sufficient precision to make the question finally tractable.

Provisional chapter structure:

1. **The View from Inside** — What a conscious agent at level ω of the hypercomputational fibration can and cannot know about its own substrate; the first-person / third-person interface as a theorem rather than a mystery.

2. **The Recapitulation** — Origin of mind as origin of life one octave up: the replication of fixed points, the emergence of self-models, and the evolutionary pressure toward accurate internal simulation.

3. **The Blind Spot, Resolved** — Returning to the blind spot identified in the Pledge: why the computational substrate of mind is invisible to the processes running on it, and why this invisibility is not a bug but a feature.

4. **Finding Mind** — The synthesis: what it means to find mind in the structure of computation, and what that finding demands of us — scientifically, philosophically, and ethically.

The Prestige is forthcoming. Readers wishing to follow its development are invited to the author's Substack at `https://f1r3fly.substack.com`.